

The stator phase voltages are computed from the stator equations in the synchronous reference frames by using the transformation on the synchronous d and q axes voltages; these are given by

$$v_{qs}^e = (R_s + L_s p) i_{qs}^e + \omega_s L_s i_{ds}^e + L_m p i_{qr}^e + \omega_s L_m i_{dr}^e \quad (7.270)$$

$$v_{ds}^e = -\omega_s L_s i_{qs}^e + (R_r + L_r p) i_{ds}^e - \omega_s L_m i_{qr}^e + L_m p i_{dr}^e \quad (7.271)$$

$$i_{abc} = [T_{abc}^e]^{-1} i_{qdo} \quad (7.272)$$

The q and d axes stator currents are piecewise continuous, so their derivatives can be found:

$$p i_{qs}^e = -\omega_s i_{ds}^e + \delta \left(m \frac{\pi}{3} \right), m = 0, 1, 2, \dots, 6 \quad (7.273)$$

$$p i_{ds}^e = \omega_s i_{qs}^e - \delta \left(m \frac{\pi}{3} \right), m = 0, 1, 2, \dots, 6 \quad (7.274)$$

where $\delta(\theta)$ is the impulse function at discontinuities.

By substituting these into the stator equations and derivatives of the rotor quadrature and direct axis currents from the rotor equations in terms of the stator current derivatives and rotor currents, the stator q and d axes voltages are obtained as

$$v_{qs}^e = R_s i_{qs}^e + \left(L_s - \frac{L_m^2}{L_r} \right) \delta(0) + \frac{L_m^2}{L_r} \omega_r i_{ds}^e - \frac{R_r L_m}{L_r} i_{qr}^e + \omega_r L_m i_{dr}^e \quad (7.275)$$

$$v_{ds}^e = -\frac{L_m^2}{L_r} \omega_r i_{qs}^e + R_s i_{ds}^e - \left(L_s - \frac{L_m^2}{L_r} \right) \delta(0) - L_m \omega_r i_{qr}^e - \frac{R_r L_m}{L_r} i_{dr}^e \quad (7.276)$$

where the impulse functions are of same magnitude as at the initial value.

From these voltages, the phase voltages are computed, and all of the variables are shown in Figure 7.51. Note that the impulse function is multiplied by the stator transient leakage inductance, which is in terms of the stator and rotor self-inductances and the mutual inductance. The magnitude of these spikes is limited by the commutating capacitors in the inverter; hence, their values in steady state are constant. Note that these spikes occur predominantly at 6 times in the stator phase voltages corresponding to the discontinuities in the phase currents. As it is not possible to have currents with zero rise and fall times, the voltage spikes are more of a function of the circuit and induction motor parameters. These spikes have to be taken into account for rating the inverter devices and commutating capacitors and for specifying the insulation of the induction motor.

The rotor flux linkages are constants except at the 60th-degree instants, where they seem to have very small changes. The rotor currents have a phase displacement of 180 degrees from their respective stator counterparts in synchronous reference

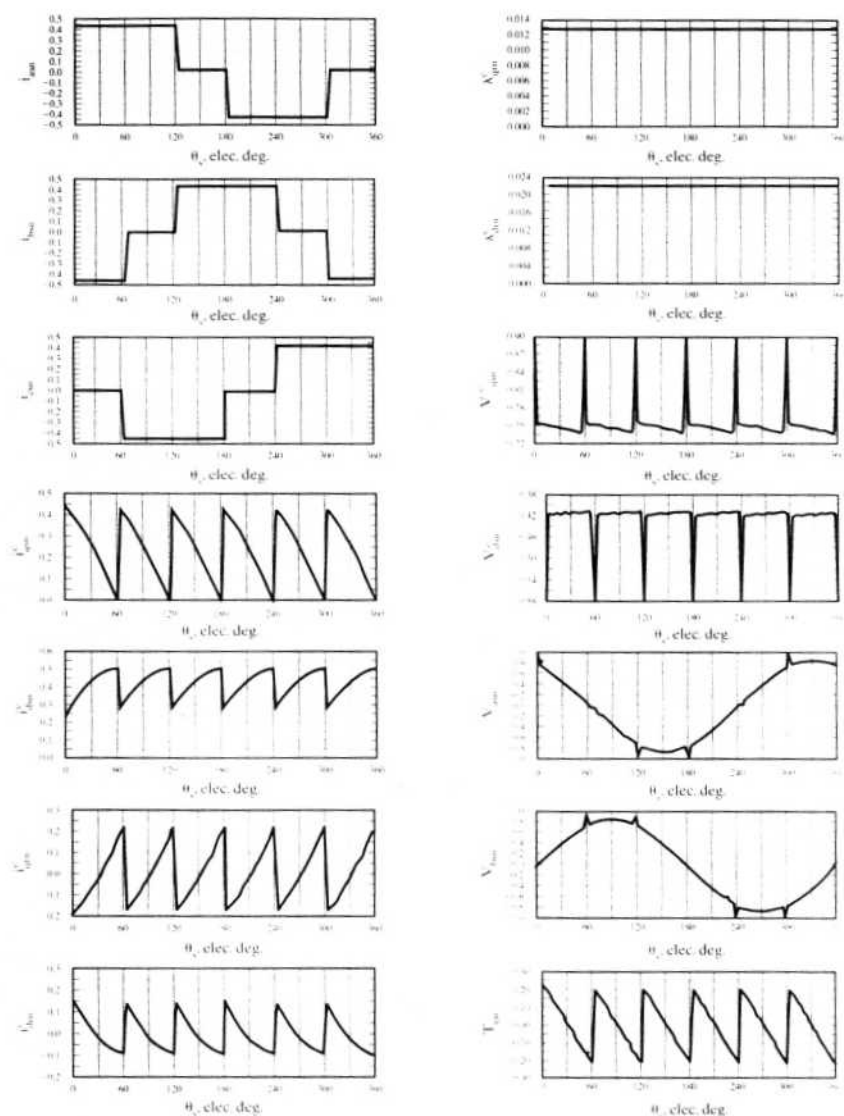


Figure 7.51(i) Steady-state operation of the CSIM at no-load, and its various waveforms in p.u.

frames but maintain the same periodicity. The q and d axes voltages have the spikes at the commutating current intervals but of opposite polarity, and the phase voltages are almost sinusoidal with superposed voltage spikes due to the current discontinuities. The electromagnetic torque has a large sixth-harmonic pulsation, as is seen from the figure, and this is one of the significant disadvantages in the six-step CSIM drive system.

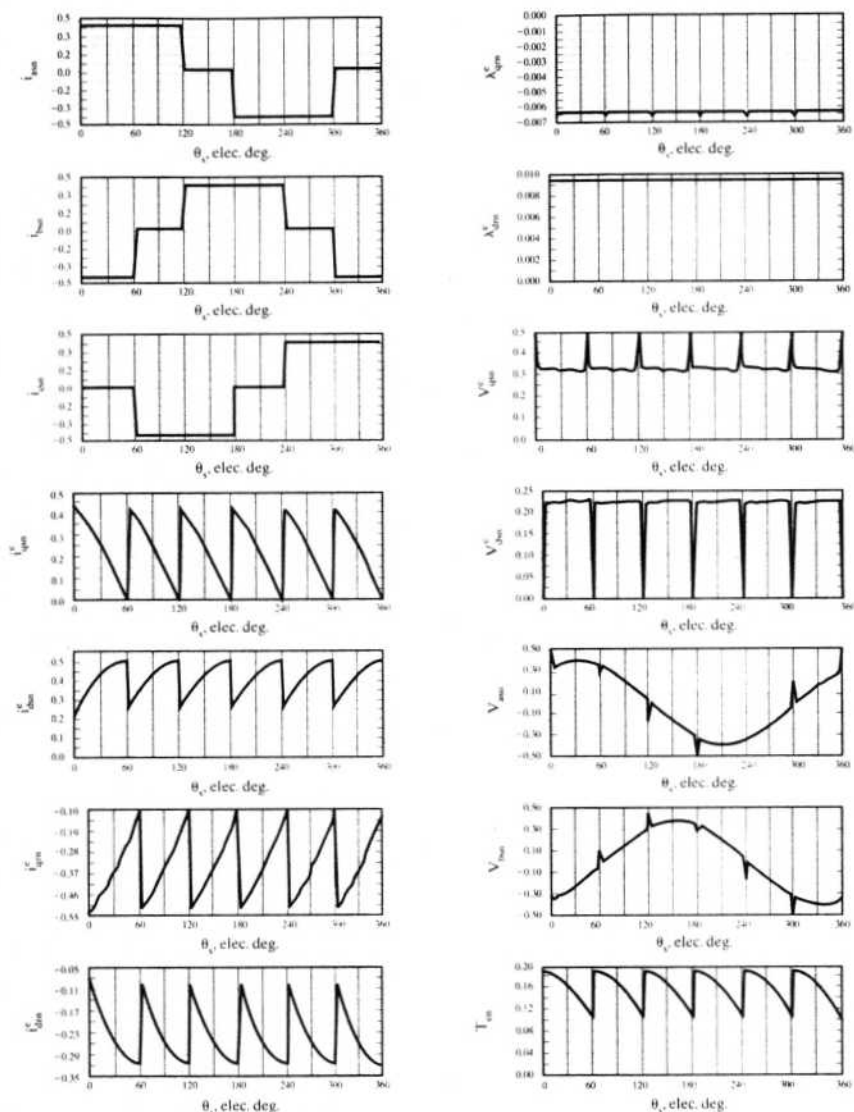


Figure 7.51(ii) Steady-state waveforms of the CSIM drive with load ($\omega_{s1} = 6.65$ rad/sec, $I_{dc} = 6.5$ A, $T_{co} = 0.159$ p.u.)

7.5.5 Closed-Loop CSIM Drive System

The schematic of the CSIM drive system is shown in Figure 7.52. The induction motor is supplied by an autosequentially commutated inverter (ASCI) with step currents. The control inputs to the inverter are current and frequency commands. The inverter control circuit enforces the stator frequency instantaneously by switching of the inverter power devices. The inverter input current is regulated by the

machine phase voltages. The reversal of the voltage at the inverter input results in instantaneous increase in dc link current, which results in a negative current error. This negative current error produces a negative control voltage, making the controlled rectifier produce a negative voltage across its output so as to oppose the inverter input voltage and thereby maintain the dc link current at its commanded value. The reversal of the converter voltage is made possible by increasing the triggering-angle delay to the converter, resulting in the operation of the converter in its inverter mode. During this time, note that the converter output voltage is negative but its output current is positive, thus producing a negative power, implying that power from the machine is returned to the ac mains via the dc link.

7.5.6 Dynamic Simulation of the Closed-loop CSIM Drive System

The dynamic simulation of the CSIM drive system is considered under the following assumptions:

- (i) The stator currents have negligible rise and fall times.
- (ii) The voltage spikes due to the step currents are constant.
- (iii) The time delay in the inverter is neglected, even though it could have a substantial effect at high speeds.

The voltage spikes are a function of the machine leakage inductances and commutating capacitors and hence are determined by the machine and inverter designers. The spikes do not, in general, affect the electromagnetic torque. The assumption that they are of fixed magnitude will not affect the dynamic simulation results, except in the case of the selection and specification of the inverter and controlled-rectifier switch ratings.

Various subsystem models are derived subsequently, and then they are integrated, to obtain the system equations with the limiter constraints in the speed and current controllers.

Stator Currents: The reference frames considered are synchronous and the stator currents are generalized as in the steady-state modeling section:

$$i_{qs}^e = g_c i_{dc} \quad (7.277)$$

$$i_{ds}^e = g_s i_{dc} \quad (7.278)$$

where

$$g_c = \frac{2}{\sqrt{3}} \cos\left(\theta_s + \frac{\pi}{6} - \{k-1\}\frac{\pi}{3}\right) \quad (7.279)$$

$$g_s = \frac{2}{\sqrt{3}} \sin\left(\theta_s + \frac{\pi}{6} - \{k-1\}\frac{\pi}{3}\right) \quad (7.280)$$

where

$$k = 1, 2, \dots, 6, 1, 2, \dots \quad (7.281)$$

defines the corresponding 60-degree intervals of stator phase angle and i_{dc} is the inverter input current. The inverter input current is the dc link current, unlike in the case of the VSIM drive systems.

The derivatives of the stator currents are obtained here for subsequent use in the simulation. They are the same as those derived in the direct steady-state evaluation section, except that they have additional terms involving the derivative of the inverter input current:

$$p i_{qs}^e = g_c p i_{dc} + i_{dc} p g_c + \delta \left(\{k - 1\} \frac{\pi}{3} \right) = g_c p i_{dc} + i_{dc} \{-\omega_s g_s\} + \delta \left(\{k - 1\} \frac{\pi}{3} \right) \quad (7.282)$$

Similarly, the derivative of the d axis current is

$$p i_{ds}^e = g_s p i_{dc} + \omega_s g_c i_{dc} - \delta \left(\{k - 1\} \frac{\pi}{3} \right) \quad (7.283)$$

where δ is the impulse contributing to the voltage spikes at the instants of current commutation occurring at stator phase angles every 60 electrical degrees.

DC Link and Power Equivalence The input inverter power is equal to the power input to the machine (neglecting losses in the inverter), and it is represented as

$$v_i i_{dc} = \frac{3}{2} [v_{qs}^e i_{qs}^e + v_{ds}^e i_{ds}^e] + v_o i_o \quad (7.284)$$

The zero-sequence current is zero, because the stator currents are balanced three-phase inputs and hence the zero-sequence power becomes zero. Substituting for the d and q axes currents in terms of the inverter input current yields

$$v_i = \frac{3}{2} [g_c v_{qs}^e + g_s v_{ds}^e] \quad (7.285)$$

This shows the dependence of the inverter input voltage on the d and q axes machine voltages; it compacts the two stator equations into one equation in terms of the inverter input voltage. The next step in the modeling is to find the stator d and q axes voltages, which involve stator and rotor currents and their derivatives. Rotor flux linkages are employed, as in the direct steady-state evaluation, because they are continuous even in the face of the discontinuous stator currents. From these variables, the stator voltages are derived.

Rotor Equations The rotor equations in terms of the rotor flux linkages are

$$p \lambda_{qr}^e = -\frac{R_r}{L_r} \lambda_{qr}^e - \omega_{sl} \lambda_{dr}^e + \frac{R_r L_m}{L_r} i_{qs}^e \quad (7.286)$$

$$p \lambda_{dr}^e = -\frac{R_r}{L_r} \lambda_{dr}^e + \omega_{sl} \lambda_{qr}^e + \frac{R_r L_m}{L_r} i_{ds}^e \quad (7.287)$$

where λ_{qr}^e and λ_{dr}^e are as defined in equations (7.247) and (7.248).

Stator Voltages The stator q and d axes voltages are derived in terms of the rotor flux linkages and stator currents as

$$v_{qs}^e = (R_s + L_s p)i_{qs}^e + \omega_s L_s i_{ds}^e + L_m p i_{qr}^e + \omega_s L_m i_{dr}^e \quad (7.288)$$

Substituting for the rotor currents in terms of the rotor flux linkages and stator currents from the equations (7.251) and (7.252) results in the q axis stator voltage:

$$v_{qs}^e = R_s i_{qs}^e + a_1 p i_{qs}^e + a_1 \omega_s i_{ds}^e + \frac{L_m}{L_r} p \lambda_{qr}^e + \frac{L_m}{L_r} \omega_s \lambda_{dr}^e \quad (7.289)$$

where

$$a_1 = L_s - \frac{L_m^2}{L_r} \quad (7.290)$$

The stator currents are substituted for in terms of the inverter input current and their derivatives, yielding

$$v_{qs}^e = R_s g_c i_{dc} + a_1 g_c p i_{dc} + \frac{L_m}{L_r} p \lambda_{qr}^e + \frac{L_m}{L_r} \omega_s \lambda_{dr}^e + a_1 \delta \left(\{k - 1\} \frac{\pi}{3} \right) \quad (7.291)$$

Similarly, the d axis stator voltage is

$$v_{ds}^e = R_s g_s i_{dc} + a_1 g_s p i_{dc} + \frac{L_m}{L_r} p \lambda_{dr}^e - \frac{L_m}{L_r} \omega_s \lambda_{qr}^e - a_1 \delta \left(\{k - 1\} \frac{\pi}{3} \right) \quad (7.292)$$

These two equations are substituted into the inverter input voltage equation for subsequent use in the dc link equation, given by

$$L_r p i_{dc} = v_r - v_i \quad (7.293)$$

where v_r is the product of the controlled-rectifier gain and v_c^* . Its real-time modeling is described in Chapter 3. Noting that v_i has as derivative the inverter input current; rearranging this equation in terms of the system variables results in the following state equation of the inverter input current:

$$p i_{dc} = \frac{-1}{2a_1 + L_r} \left[2R_s i_{dc} + 1.5 \frac{L_m}{L_r} \{g_c p \lambda_{qr}^e + g_s p \lambda_{dr}^e\} + 1.5 \frac{L_m}{L_r} \omega_s \{g_c \lambda_{dr}^e - g_s \lambda_{qr}^e\} + 1.5 a_1 (g_c - g_s) \delta \left(\{k - 1\} \frac{\pi}{3} \right) - v_r \right] \quad (7.294)$$

Load Interaction The machine and load interaction is given by the equation with inertia and friction constant:

$$J p \omega_m + B \omega_m = T_e - T_l \quad (7.295)$$

where the electromagnetic torque is given by

$$T_e = \frac{3}{2} \frac{P}{2} L_m [i_{qs}^e i_{dr}^e - i_{ds}^e i_{qr}^e] \quad (7.296)$$

which, in terms of the rotor flux linkages, is given by

$$T_e = \frac{3}{2} \frac{P}{L_r} \frac{L_m}{L_r} [\lambda_{dr}^e g_c - \lambda_{qr}^e g_s] i_{dc} \quad (7.297)$$

In terms of the rotor electrical speed, equation (7.295) becomes, by the multiplication of the pole-pairs,

$$p\omega_r = -\frac{B}{J}\omega_r + \frac{1}{J} \frac{3}{2} \frac{P^2}{2} \frac{L_m}{L_r} \{\lambda_{dr}^e g_c - \lambda_{qr}^e g_s\} i_{dc} - \frac{1}{J} \frac{P}{2} T \quad (7.298)$$

Equations (7.291) to (7.297) describe the machine, inverter, and load whereas the stator frequency and rectifier output-voltage variables are obtained from the controller structure consisting of the reference speed, speed-controller and current-controller constants, and slip-speed-to-inverter current-command transfer block. These are obtained as follows.

Frequency Command The stator frequency is generated by the rotor speed loop and the slip-speed command, generated by the speed error between the reference and actual speed. The tachogenerator, with its conditioning circuit, is modeled as a gain block, but, in practice, it will contain a first-order filter. The tachogenerator output is proportional to the rotor speed, and the slip-speed command is obtained by passing the speed error through a proportional and integral controller with the gains of K_{ps} and K_{is} , respectively. The slip-speed signal is limited by the fact that the operation of the induction motor has to be constrained within the high-efficiency region of the slip torque, which corresponds to the statically stable portion of the voltage-source-fed induction-motor characteristics. The equations of relevance are the following:

$$v_{tg} = K_{tg} \omega_r \quad (7.299)$$

$$v_{sl}^* = K_{ps}(v^* - v_{tg}) + K_{is} \int (v^* - v_{tg}) dt \quad (7.300)$$

where v^* is the external speed reference.

Then, the stator frequency command is given by

$$f_s^* = \frac{v_{tg} + v_{sl}^*}{2\pi K_{tg}} \quad (7.301)$$

The slip-speed command is then obtained by dividing equation (7.300) by K_{tg} . The next step is to generate the command signal for the controlled rectifier; this step requires the generation of the inverter input-current command.

DC-Link Current Command From the induction motor steady-state equivalent circuit, the stator rms current is written in terms of the magnetizing current as

$$I_s = I_m + I_r = I_m + \frac{j\omega_s L_m I_m}{\frac{R_r}{s} + j\omega_s L_{lr}} = I_m \left[1 + \frac{j s \omega_s L_m}{R_r + j s \omega_s L_{lr}} \right] \quad (7.302)$$

Note that $\omega_{sl} = s\omega_s$; hence,

$$I_s = I_m \left[\frac{R_r + j\omega_{sl}L_r}{R_r + j\omega_{sl}L_{lr}} \right] \quad (7.303)$$

The relationship between the fundamental stator phase and dc link currents is

$$i_{dc} = \frac{I_s}{0.778} = 1.285I_s \quad (7.304)$$

Then the inverter input-current command is obtained by combining the last two expressions:

$$i_{dc}^* = 1.285I_m \left| \frac{R_r + j\omega_{sl}L_r}{R_r + j\omega_{sl}L_{lr}} \right| = 1.285I_m \frac{\sqrt{R_r^2 + (\omega_{sl}L_r)^2}}{\sqrt{R_r^2 + (\omega_{sl}L_{lr})^2}} \quad (7.305)$$

Thus, the link-current command is a function of only the slip-speed command, assuming that the magnetizing current is kept constant. An on-line computation of the inverter input-current command would facilitate the variation of the magnetizing current as a function of slip speed, to optimize the efficiency in steady-state operation.

Controlled-Rectifier Voltage Command The inverter current command is compared to the measured value of the inverter input current to produce an error signal, which is amplified through a proportional-plus-integral current controller whose gains are K_{pi} and K_{ii} , respectively. The output of the current controller is limited, to provide the safe operation of the converter during inversion. The output of the current controller actuates the controlled rectifier to provide a proportional output voltage, v_r , to generate the inverter input current to match its command. The control-voltage signal and the rectifier output voltage are given by

$$v_c^* = K_{pi}(i_{dc}^* - i_{dc})H_c + K_{ii} \int \{i_{dc}^* - i_{dc}\}H_c dt \quad (7.306)$$

where H_c is the current transducer gain. Note that the inverter input current command and slip speed were derived above, and they should be substituted into this expression to obtain the voltage-command signal.

System Integration and Equations The equations of the various subsystems can now be assembled in state-space form for simulation. The following definitions are made to get them into state-space form:

$$\begin{aligned} x_1 &= i_{dc} \\ x_2 &= \lambda_{qr}^c \\ x_3 &= \lambda_{dr}^c \\ x_4 &= \omega_r \\ x_5 &= \int (v^* - K_{te}\omega_r) dt \\ x_6 &= \int H_c(i_{dc}^* - i_{dc}) dt \end{aligned} \quad (7.307)$$

The system differential equations in terms of the states defined are

$$\begin{aligned}
 p x_1 &= \frac{-1}{2a_1 + L_r} \left[2R_s x_1 + 1.5 \frac{L_m}{L_r} \{g_c p x_2 + g_s p x_3\} \right. \\
 &\quad \left. + 1.5 \frac{L_m}{L_r} \{g_c x_3 - g_s x_2\} + 1.5a_1(g_c - g_s) \delta \left(\{k-1\} \frac{\pi}{3} \right) - v_r \right] \\
 p x_2 &= -\frac{R_r}{L_r} x_2 - \omega_{sl} x_3 + \frac{R_r L_m}{L_r} g_c x_1 \\
 p x_3 &= \omega_{sl} x_2 - \frac{R_r}{L_r} x_3 + \frac{R_r L_m}{L_r} g_s x_1 \\
 p x_4 &= \frac{13}{J2} \left(\frac{P}{2} \right)^2 \frac{L_m}{L_r} \{g_c x_3 - g_s x_2\} x_1 - \frac{B}{J} x_4 - \frac{P}{2J} T_l \\
 p x_5 &= v^* - K_{tg} x_4 \\
 p x_6 &= H_c(i_{dc}^* - x_1)
 \end{aligned} \tag{7.308}$$

Note that these equations have only the inverter input-current command, slip frequency, stator frequency, and external speed reference as external inputs; they are described by the following equations in terms of the states as

$$\omega_{sl} = \frac{v_{sl}^*}{K_{tg}} = \frac{K_{is} x_5 + K_{ps}(V^* - K_{tg} x_4)}{K_{tg}} = \frac{K_{is} x_5 + K_{ps} p x_5}{K_{tg}} \tag{7.309}$$

$$\omega_s = \omega_r + \omega_{sl} = x_4 + \frac{K_{is} x_5 + K_{ps}(V^* - K_{tg} x_4)}{K_{tg}} = x_4 + \frac{K_{is} x_5 + K_{ps} p x_5}{K_{tg}} \tag{7.310}$$

The inverter input-current command is derived from equations (7.305) and (7.309) as

$$i_{dc}^* = 1.285 I_m \frac{\sqrt{(R_r K_{tg})^2 + L_r^2 \{K_{is} x_5 + K_{ps} p x_5\}^2}}{\sqrt{(R_r K_{tg})^2 + L_r^2 \{K_{is} x_5 + K_{ps} p x_5\}^2}} \tag{7.311}$$

The voltage-command signal is given by

$$v_c^* = K_{pi} p x_6 + K_{ii} x_6 \tag{7.312}$$

The slip-speed and voltage-command signals are limited; their limits are given as

$$-V_{cm} < v_c^* < +V_{cm} \tag{7.313}$$

$$-V_{sm} < v_{sl}^* < +V_{sm} \tag{7.314}$$

These limiters are applied to the relevant state variables and their derivatives during computation.

A sample CSIM drive system simulation is shown in Figure 7.53 for the induction motor whose parameters are given in Example 5.1; the system parameters are given below.

$V_{ll} = 240$ V, $L_r = 0.1$ H, $K_{ps} = 10$, $K_{is} = 1$, $K_{pi} = 10$, $K_{ii} = 0.1$, speed command = 1 p.u.

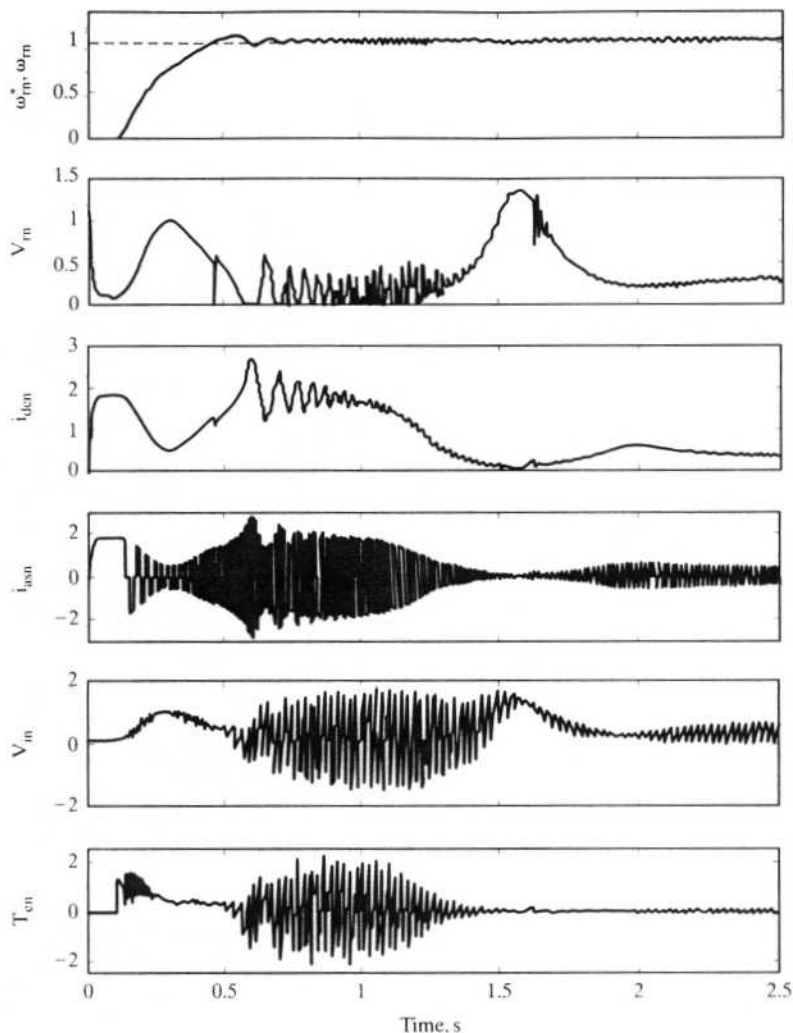


Figure 7.53(i) Normalized transient response for 1-p.u.-step speed command, from standstill

A one-p.u.-step command is given to the motor drive system. The transient responses are shown in Figure 7.53(i). The machine takes approximately 0.1 s to raise the rotor flux to rated value. Until then, the stator current is used entirely to set up the flux. The machine is accelerated to 1-p.u. speed in 0.4 s thereafter. The machine has a 0.2 p.u. load; only friction has to be overcome. To reach steady state, the drive goes through a severe oscillation. This is one of the characteristic disadvantages of the ASCI drive with the adopted control strategy. Almost an additional 0.2 s is taken to reach steady state. The steady-state waveforms are shown in Figure 7.53(ii). The cusps in the front-end converter's output are due to

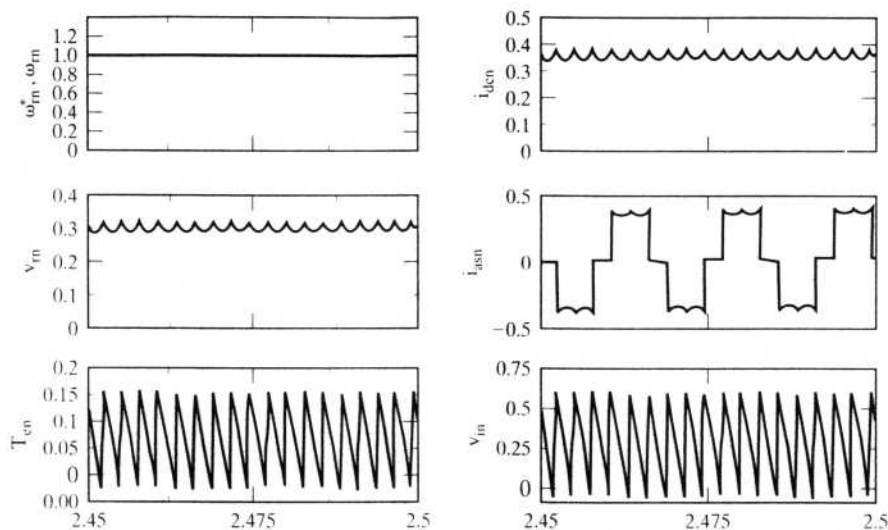


Figure 7.53(ii) Normalized steady-state responses from Figure 7.54(i).

the v_c^* variations, and the output of the converter is modeled as the product of the v_c^* and gain of the converter. These cusps in v_{in} , and hence in i_{dcn} , are transmitted to the phase currents and are shown in the phase a current. The torque shows large pulsations at six times the fundamental frequency, and so does the inverter input voltage.

7.6 APPLICATIONS

Voltage-source induction motor drives are used as general-purpose drives in such applications as fans, pumps, packaging, conveyor, hand tools, and appliances. These applications do not require positioning capabilities, and precise torque control in the 0-to-5-Hz range is not critical.

Use of software-based controllers and the increasing use of modern control theory in the synthesis of the controllers has resulted recently in a far wider control range with these drives. Because of this factor, these drives have made immense strides in the marketplace by enlarging their share.

The hard-current source drives are popular in limited high-power (>100 hp) applications, such as paper mills and pulp, sugar, and rubber centrifuges, where reliability and ruggedness are major considerations. The soft current-source induction motor drives have eliminated the disadvantages of the hard current-source drives, and, with higher controllers such as vector controllers, have moved into demanding precision applications such as machine-tool servo and spindle drives.

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7.8 DISCUSSION QUESTIONS

1. Why is it that the variable-frequency requirement is always accompanied by a variable-voltage requirement in ac machines?
2. Inverters provide variable voltage and variable frequency. Will they provide current control? If so, how is it accomplished?
3. What is the greatest shortcoming in a voltage-source inverter?
4. What is the greatest shortcoming in a current-source inverter?
5. Compare voltage- and current-source inverters in terms of their reliability, control complexity, and harmonics.
6. For induction motor control, compare the suitability of the voltage- and current-source inverters in terms of their control complexity and reliability.
7. Compare the performance features of six-step and PWM control of voltage-source inverters in terms of harmonics, control complexity, and ease of implementation.
8. Compare the features of the PWM and hysteresis control of inverters, and identify, with justification, the most desirable control in most of the applications.
9. If the pulse-width modulation ratio is increased beyond one, what happens to the output voltage? If there is a side effect to this mode of operation, identify it. (Hint: Refer to a power electronics text.)
10. What is the maximum rms voltage obtainable with a six-step operation in a voltage-source inverter? DC link voltage is V_{dc} . Justify your assumptions.

11. What is the maximum rms voltage obtainable under PWM operation by using triangular carrier frequency and sine modulation signal in a voltage-source inverter? Justify your assumptions.
12. Discuss the differences between a hard (ASCI) and a soft (PWM-based) current-source inverter.
13. If the line-to-line voltages are unbalanced in the inverter output, are the phase voltages determined by equations (7.8), (7.9), and (7.10)?
14. An inverter has to be rated in apparent power rather than in real power. Explain why.
15. Discuss the effects of unbalanced voltages on the operation of the induction motor.
16. How are unbalanced voltages detected and corrected in the inverter?
17. Discuss the merits of the volts/Hz control strategy.
18. Draw the control and power schematics of the volts/Hz drive that uses a PWM inverter.
19. Discuss the difference between the six-step and the PWM inverter with volts/Hz control for an induction motor.
20. Offset voltage is a variable dependent on the stator impedance and stator current. Discuss a method for an accurate implementation of offset-voltage control in volts/Hz-controlled induction motor drive.
21. Discuss the parameter sensitivity of the offset voltage that is due to stator resistance in volts/Hz controller.
22. Dynamic response of a volts/Hz-controlled induction motor drive is characterized by large starting stator currents. Explain why.
23. How can this starting-current magnitude be controlled?
24. Discuss the cause of starting-current transient in a volts/Hz-controlled induction motor.
25. Explain the distinct advantages of direct-steady-state-performance evaluation of the induction motor compared to the method using an equivalent circuit.
26. Discuss the important advantage of slip-speed control over the volts/Hz control strategy.
27. Can the direct-steady-state evaluation method be used for predicting the performance of the constant-slip-speed-controlled induction motor drive?
28. Compare the principle of operation of constant-air gap-flux-linkages control as against that of the volts/Hz and constant-slip-speed controls.
29. Implementation of constant air gap flux requires stator current control. Justify this statement.
30. Will constant air gap flux be maintained by using the drive schematic shown in Figure 7.30, even during dynamic conditions?
31. Discuss the merits and demerits of the multistep and PWM-inverter control.
32. How will the PWM voltages, as compared to six-step voltages, affect the torque pulsation both in magnitude and in frequency?
33. PWM voltages eliminate lower-order harmonics but increase higher-order harmonics. How does this fact affect motor losses?
34. Is PWM applicable to hard current-source inverters also?
35. Which of the following will have a larger sixth-harmonic torque pulsation: (a) six-step voltage source, (b) six-step current source? Induction motor is operating at rated current/rated voltage and at rated speed.
36. The load has no impact on peak-to-peak value of sixth-harmonic torque pulsation in a voltage-source-inverter-fed induction motor. Why is that?

37. Load decreases the severity of the sixth-harmonic torque pulsation in the current-source-inverter-fed induction motor drive, as is seen from Figures 7.51(i) and 7.51(ii). Explain the reason for this.
38. The inner speed loop in the induction motor drive acts to keep the machine in a statically stable region and therefore in synchronous lock. Without a speed sensor, how can such a loop be implemented?

7.9 EXERCISE PROBLEMS

1. Consider the machine parameters described in problem 7, section 6.6. Instead of using phase control for the induction motor drive, variable-frequency control with a constant-volts/Hz strategy is used. Find the efficiency of the motor drive with this drive system for the same application. [Hint: From the operating speed, torque, and volts/Hz relationship, the slip is derived through the solution of a quadratic equation.]
2. Assume that the price of energy is \$0.05/kWh (kWh — kilowatthour). The fan considered in problem 1 is running eight hours per day for 356 days in a year. The price of the phase controller is \$2000 and the price of the variable-frequency drive with constant-volts/Hz control is \$3500. It is planned to have one of the variable-speed drives installed for the problem studied. If the variable-frequency drive is to be chosen instead of the phase controller, what is the payback period for the difference in the initial investment involved? Consider that the capital is acquired interest-free for this problem. (In practice, there is an interest for the loan.)
3. Consider the 5-hp induction motor described in Example 5.1. Find its slip speed, stator and rotor current magnitudes, efficiency, and power factor when it is delivering 12 Nm air gap torque for two control strategies:
 - (i) volts/Hz control
 - (ii) slip-speed control
4. A 100-hp induction motor is to drive a fan in a warehouse. It is found that the air volume has to be reduced during the off-peak business hours. That corresponds to an operational speed of 65% of the rated speed. The motor details are given next:

100 hp, 460 V, 3 phase, 60 Hz, 2 poles, star connected squirrel cage

induction motor, $R_s = 0.05 \Omega$, $R_r = 0.0288 \Omega$, $X_m = 10.5 \Omega$,

$X_{ls} = 0.3 \Omega$, $X_{lr} = 0.405 \Omega$, Rated speed = 3555 rpm.

Find the efficiency of the motor drive when it is operated from a six-step inverter with a volts/Hz control strategy for the following two cases:

- (i) with no offset voltage
 - (ii) with an offset voltage of 4.7 V
- The fan has a rated torque requirement at rated speed.
5. Prove that, in an air gap-flux-controlled induction motor drive,
 - (i) air gap torque is a function of only magnetizing current and slip speed, because these two can be independently varied in this drive system, and
 - (ii) the maximum torque occurs at the slip frequency $\omega_{sl} = \frac{R_r}{L_{lr}}$.
 6. In a volts/Hz-controlled induction motor, only applied voltage and stator frequency are control variables.

- (i) Determine the slip speed at which the air gap torque is maximum for this motor drive. The magnetizing impedance is considered to be very large compared to stator and rotor impedances.
 - (ii) Compare the efficiency of this drive at maximum torque to the air gap-flux-controlled induction motor drive.
7. Consider the problem in Example 7.5. Include the effects of stator impedance, and recalculate the magnitude of the sixth-harmonic torque pulsation.
 8. Find the range of rotor speed for which the air gap power is constant for the machine considered in Example 7.8. Use the formula developed in the text. Also, calculate the actual range of speed, with the consideration that the stator phase voltage and current are constrained to be equal to or less than 1 p.u..
 9. The slip frequency is set as the product of the rated slip and stator frequency. Flux weakening is introduced beyond rated frequency with

$$\omega_s^* = \omega_r + \omega_{sl}^* = \omega_r + s_r \omega_s^*$$

resulting in $\omega_s^* = \frac{\omega_r}{1 - s_r}$. Find the range of speed for which the air gap power is at rated value for this flux-weakening strategy.

10. A six-step current-source inverter is feeding the induction motor whose details are given in Example 5.1. Calculate the magnitude of the sixth-harmonic torque pulsation when the machine is delivering rated torque at a stator frequency of 10 Hz.
11. Simulate a PWM-based, volts/Hz-controlled induction motor drive with a stator current limit.
12. Simulate the PWM current-source-based induction motor drive with slip-speed limit. Comment on the salient differences between this simulation and the ASCI drive results given in Figure 7.53.
13. Develop an algorithm to evaluate the losses in a machine supplied from a PWM voltage-source inverter.
14. The current-command generator is sensitive to rotor-resistance variation in current-source drives. Develop a method that is not on-line-computation-based to generate the current command when the rotor-resistance estimate is available as a signal.

CHAPTER 8

Vector-Controlled Induction Motor Drives

8.1 INTRODUCTION

The various control strategies for the control of the inverter-fed induction motor have provided good steady-state but poor dynamic response. From the traces of the dynamic responses, the cause of such poor dynamic response is found to be that the air gap flux linkages deviate from their set values. The deviation is not only in magnitude but also in phase. The variations in the flux linkages have to be controlled by the magnitude and frequency of the stator and rotor phase currents and their instantaneous phases. So far, the control strategies have utilized the stator phase current magnitude and frequency and not their phases. This resulted in the deviation of the phase and magnitudes of the air gap flux linkages from their set values.

The oscillations in the air gap flux linkages result in oscillations in electromagnetic torque and, if left unchecked, reflect as speed oscillations. This is undesirable in many high-performance applications, such as in robotic actuators, centrifuges, servos, process drives, and metal-rolling mills, where high precision, fast positioning, or speed control are required. Such requirement will not be met with the sluggishness of control due to the flux oscillations. Further, air gap flux variations result in large excursions of stator currents, requiring large peak converter and inverter ratings to meet the dynamics. An enhancement of peak inverter rating increases cost and reduces the competitive edge of ac drives in the marketplace, in spite of excellent advantages of the ac drives over dc drives.

Separately-excited dc drives are simpler in control because they independently control flux, which, when maintained constant, contributes to an independent control of torque. This is made possible with separate control of field and armature currents which, in turn, control the field flux and the torque independently. Moreover, the dc motor control requires only the control of the field or armature current magnitudes, providing a simplicity not possible with ac machine control. By

contrast, ac induction motor drives require a coordinated control of stator current magnitudes, frequencies, and their phases, making it a complex control. As with the dc drives, independent control of the flux and torque is possible in ac drives. The stator current phasor can be resolved, say, along the rotor flux linkages, and the component along the rotor flux linkages is the field-producing current, but this requires the position of the rotor flux linkages at every instant; note that this is dynamic, unlike in the dc machine. If this is available, then the control of ac machines is very similar to that of separately-excited dc machines. The requirement of phase, frequency, and magnitude control of the currents and hence of the flux phasor is made possible by inverter control. The control is achieved in field coordinates (hence the name of this control strategy, *field-oriented control*); sometimes it is known as *vector control*, because it relates to the phasor control of the rotor flux linkages.

Vector control made the ac drives equivalent to dc drives in the independent control of flux and torque and superior to them in their dynamic performance. These developments positioned the ac drives for high-performance applications, hitherto reserved for separately-excited dc motor drives. This chapter describes the basic principles, classifications, derivation, modeling, analysis, and design of vector-control schemes. The parameter sensitivity of vector-control schemes is analyzed, and methods for its compensations are described. The design of the speed controller for a vector controller is systematically derived by using symmetric optimum technique.

8.2 PRINCIPLE OF VECTOR CONTROL

To explain the principle of vector control, an assumption is made that the position of the rotor flux linkages phasor, λ_r , is known. λ_r is at θ_r from a stationary reference, θ_r is referred to as field angle hereafter, and the three stator currents can be transformed into q and d axes currents in the synchronous reference frames by using the transformation

$$\begin{bmatrix} i_{qs}^e \\ i_{ds}^e \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \sin \theta_r & \sin \left(\theta_r - \frac{2\pi}{3} \right) & \sin \left(\theta_r + \frac{2\pi}{3} \right) \\ \cos \theta_r & \cos \left(\theta_r - \frac{2\pi}{3} \right) & \cos \left(\theta_r + \frac{2\pi}{3} \right) \end{bmatrix} \begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix} \quad (8.1)$$

from which the stator current phasor, i_s , is derived as

$$i_s = \sqrt{(i_{qs}^e)^2 + (i_{ds}^e)^2} \quad (8.2)$$

and the stator phasor angle is

$$\theta_s = \tan^{-1} \left\{ \frac{i_{qs}^e}{i_{ds}^e} \right\} \quad (8.3)$$

where i_{qs}^e and i_{ds}^e are the q and d axes currents in the synchronous reference frames that are obtained by projecting the stator current phasor on the q and d axes, respectively. That the current phasor magnitude remains the same regardless of the reference frame chosen to view it is evident from Figure 8.1. The current phasor i_s

Crucial to the implementation of vector control, then, is the acquiring of the instantaneous rotor flux phasor position, θ_r . This field angle can be written as

$$\theta_r = \theta_r + \theta_{sl} \quad (8.6)$$

where θ_r is the rotor position and θ_{sl} is the slip angle. In terms of the speeds and time, the field angle is written as

$$\theta_r = \int (\omega_r + \omega_{sl}) dt = \int \omega_s dt \quad (8.7)$$

Vector-control schemes are classified according to how the field angle is acquired. If the field angle is calculated by using terminal voltages and currents or Hall sensors or flux-sensing windings, then it is known as *direct vector control*. The field angle can also be obtained by using rotor position measurement and partial estimation with only machine parameters but not any other variables, such as voltages or currents; using this field angle leads to a class of control schemes known as *indirect vector control*. The direct and indirect vector-controlled schemes are explained in the following sections.

Vector control is summarized by the following algorithm.

- (i) Obtain the field angle.
- (ii) Calculate the flux-producing component of current, i_r^* , for a required rotor flux linkage λ_r^* . By controlling only this field current, the rotor flux linkages are controlled. It is very similar to the separately-excited dc machine, in that the field current controls the field flux; the armature current has no impact on it.
- (iii) From λ_r^* and the required T_e^* , calculate the torque-producing component of stator current, i_T^* . Controlling the torque-producing component current when the rotor flux linkages phasor is constant gives an independent control of electromagnetic torque. It is very similar to the case of the armature current's controlling the electromagnetic torque in a separately-excited dc machine with the field current maintained constant. Steps (ii) and (iii) enable a complete decoupling of flux- from torque-producing channels in the induction machine.
- (iv) Calculate the stator-current phasor magnitude, i_s^* , from the vector sum of i_T^* and i_r^* .
- (v) Calculate torque angle from the flux- and torque-producing components of the stator-current commands, $\theta_T = \tan^{-1} \frac{i_T^*}{i_r^*}$.
- (vi) Add θ_T and θ_r to obtain the stator current phasor angle, θ_s .
- (vii) By using the stator-current phasor angle and its magnitude, θ_s and i_s^* , the required stator-current commands are found by going through the *qdo* transformation to *abc* variables:

$$\begin{aligned} i_{as}^* &= i_s^* \sin \theta_s \\ i_{bs}^* &= i_s^* \sin \left(\theta_s - \frac{2\pi}{3} \right) \\ i_{cs}^* &= i_s^* \sin \left(\theta_s + \frac{2\pi}{3} \right) \end{aligned} \quad (8.8)$$

- (viii) Synthesize these currents by using an inverter; when they are supplied to the stator of the induction motor, the commanded rotor flux linkages and torque are produced.

The constants of proportionality involved in the flux and torque will be derived in later sections. The correspondence between the separately-excited dc motor and the induction motor is complete: i_f and i_T correspond to the field and armature currents of the dc machine, respectively. Even though the induction motor does not have separate field and armature windings, finding equivalent field and armature currents as components of the stator-current phasor has resulted in the decoupling of flux- from torque-producing channels in a machine that is highly coupled. Unlike the scalar control involved in dc machines, phasor or vector control is employed in induction machines. In the dc machine, the field and armature are fixed in space by the commutator, whereas, in the induction machine, no such additional component exists to separate the field (to produce the flux) from the armature (to produce the torque) channels to the optimum space angle of 90 electrical degrees between them. In the place of the commutator, the induction machine (and for that matter any ac machine) acquires the functionality of the commutator with an inverter. The inverter controls both the magnitude of the current and its phase, allowing the machine's flux and torque channels to be decoupled by controlling precisely and injecting the flux- and torque-producing currents in the induction machine to match the required rotor flux linkages and electromagnetic torque. The phasor control of current further adds to the complexity of computation involving phase and magnitude and of transformations to orient i_f and i_T with respect to rotor flux linkages. Note that the orientation need not be on rotor flux linkages; the computations can be carried out in stator or rotor or arbitrary reference frames. Synchronous reference frames are often used for the sake of freeing the signal-processing circuits from high bandwidth requirements, as is explained elsewhere.

8.3 DIRECT VECTOR CONTROL

8.3.1 Description

A block diagram of the direct vector-control scheme with a current-source inverter is shown in Figure 8.2. The electrical rotor speed, ω_r , is compared to the reference speed, ω_r^* , and the error is amplified and limited to generate the reference (command) torque, T_e^* . The rotor flux linkages reference λ_r^* is derived from the rotor speed via an absolute-value function generator. λ_r^* is kept at 1 p.u. for from 0- to 1-p.u. rotor speed; beyond 1-p.u. speed, it is varied as a function of the rotor speed. This is to ensure that the rotor speed is extended beyond the base speed with the available dc voltage to the inverter, by weakening the rotor flux linkages, thus reducing the induced emf to lower than that of the available output voltage from the inverter. By reducing the rotor flux linkages for the same

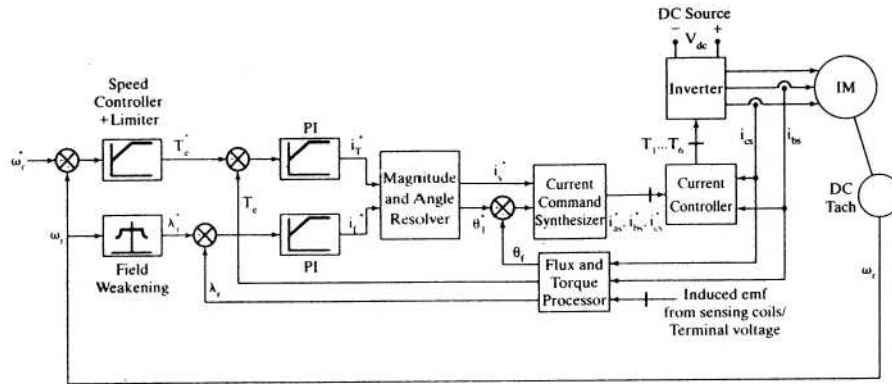


Figure 8.2 Direct vector-control scheme

torque-producing component of the stator current, the electromagnetic torque is reduced, which, in combination with the increasing rotor speed, can be controlled to produce the constant power output, say at rated value of the machine in steady state. The torque and rotor flux-linkages references are compared to the torque, T_e , and the rotor flux-linkages, λ_r , respectively. Their errors are amplified and limited to generate the reference torque- and flux-producing components of stator current, i_T^* and i_f^* , respectively. Phasor addition of i_f^* and i_T^* yields the stator-current phasor reference i_s^* , and the angle between i_T^* and i_f^* gives the torque-angle reference, θ_T^* . The sum of torque angle and field angle gives the position of the stator-current phasor, θ_s . Together with i_s^* , this generates the stator-phase current references, i_{as}^* , i_{bs}^* , and i_{cs}^* . These stator-phase-current requests are simplified by using an inverter and current feedback loops. The phase-current control loops can use one of the following switching techniques:

1. PWM
2. hysteresis
3. space-vector modulation (SVM)

Refer to Chapter 7 for a description of PWM, and hysteresis switching schemes and SVM scheme may be found in references and in some manner in this chapter. By one of these control techniques, the inverter output currents are made to correspond to the reference inputs. The feedback variables θ_r , T_e , and λ_r are obtained from the flux and torque processor block. This is the key to the direct vector-control scheme. The realization of this block will be examined in detail.

8.3.2 Flux and Torque Processor

The inputs to this block are two stator phase currents and one set of the following: either (i) terminal voltages or (ii) induced emf from the flux-sensing coils or Hall

sensors. The selection of (i) or (ii) will determine the computational algorithm. Hence, for these inputs, computational steps are developed in the following.

8.3.2.1 Case (i): Terminal voltages. By using terminal voltages, the air gap torque, flux, and field angle can be computed with either rotor or stator flux linkages. The choice has significant impact on the performance of the drive system in terms of its parameter sensitivity, as is seen from the following development.

A. Rotor-Flux-Based Calculator Two line-to-line voltages can be measured, from which the phase (line to neutral) voltages can be computed (provided that the voltages are balanced). The q and d stator voltages in the stator reference frames are obtained from the phase voltages as

$$\begin{aligned} v_{qs} &= v_{as} \\ v_{ds} &= \frac{1}{\sqrt{3}}(v_{cs} - v_{bs}) \end{aligned} \quad (8.9)$$

Similarly, the currents are obtained in the same way; these equations hold true for them, too. From the stator-reference-frame equations of the induction motor, the stator equations are

$$\begin{aligned} v_{qs} &= (R_s + L_s p)i_{qs} + L_m p i_{qr} \\ v_{ds} &= (R_s + L_s p)i_{ds} + L_m p i_{dr} \end{aligned} \quad (8.10)$$

from which the rotor currents, i_{qr} and i_{dr} , can be computed as

$$\begin{aligned} i_{qr} &= \frac{1}{L_m} \left\{ \int (v_{qs} - R_s i_{qs}) dt - L_s i_{qs} \right\} \\ i_{dr} &= \frac{1}{L_m} \left\{ \int (v_{ds} - R_s i_{ds}) dt - L_s i_{ds} \right\} \end{aligned} \quad (8.11)$$

From all the stator and rotor currents, torque, the flux and field angle can be computed as follows:

$$\begin{aligned} T_e &= \frac{3}{2} \frac{P}{2} L_m (i_{qs} i_{dr} - i_{ds} i_{qr}) \\ \lambda_{qr} &= L_r i_{dr} + L_m i_{ds} \\ \lambda_{dr} &= L_r i_{qr} + L_m i_{qs} \\ \lambda_r &= \sqrt{(\lambda_{qr})^2 + (\lambda_{dr})^2} \\ \theta_r &= \tan^{-1} \left(\frac{\lambda_{qr}}{\lambda_{dr}} \right) \end{aligned} \quad (8.12)$$

Equations (8.9) to (8.12) can be implemented with computational circuits, as shown in Figure 8.3. The same diagram could be used for the code development for implementation in a single-chip processor. Note the circuit's dependence on the motor parameters R_s , L_s , L_r , and L_m . The changes in stator resistance could be tracked

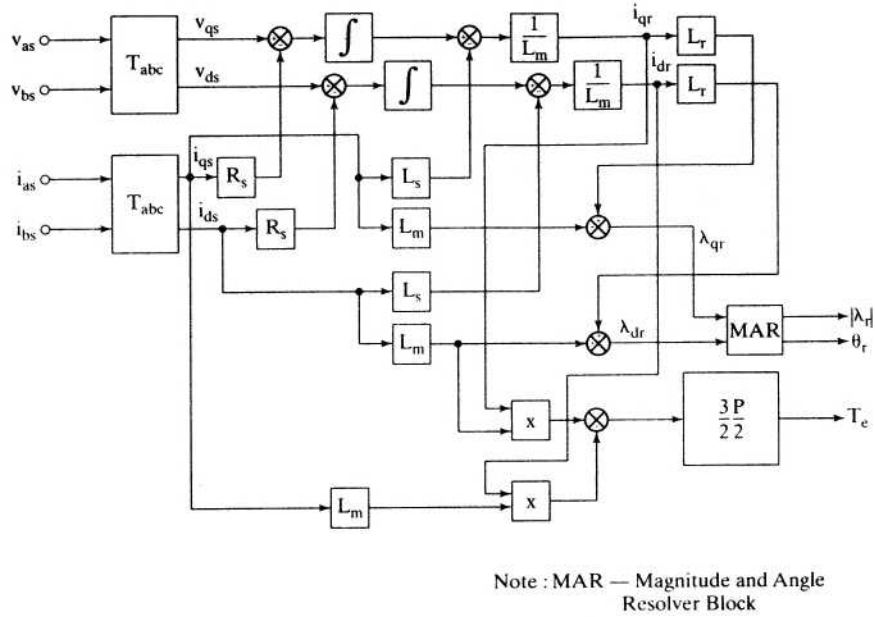


Figure 8.3 Flux and torque processor implementation

indirectly with inexpensive temperature sensors. As for the inductances, they would have no significant variations in this scheme since flux control is implemented. Hence, parameter sensitivity would not greatly plague the accuracy of the measurement and calculation of flux, torque, and field angle.

B. Stator-Flux-Based Calculator Computational steps and dependence on many motor parameters could be very much reduced by using the stator flux linkages and calculating the electromagnetic torque, using only the stator flux linkages and stator currents. Then only stator resistance is employed in the computation of the stator flux linkages, thereby removing the dependence of mutual and rotor inductances of the machine on its calculation. The steps involved are summarized briefly as follows.

$$\begin{aligned}
 \lambda_{ds} &= \int (v_{ds} - R_s i_{ds}) dt \\
 \lambda_{qs} &= \int (v_{qs} - R_s i_{qs}) dt \\
 \lambda_s &= \sqrt{(\lambda_{qs})^2 + (\lambda_{ds})^2} \angle \theta_{fs} \\
 \theta_{fs} &= \tan^{-1} \left(\frac{\lambda_{qs}}{\lambda_{ds}} \right) \\
 T_e &= \frac{3}{2} \frac{P}{2} (i_{qs} \lambda_{ds} - i_{ds} \lambda_{qs})
 \end{aligned} \tag{8.13}$$

In this case, the flux loop can be closed with the stator flux linkages instead of the rotor flux linkages. The accuracy of the computation might not be high, even though this algorithm depends only on the stator resistance rather than on many other motor parameters, as is proved by an example in the following. The sensitivity of the stator resistance variation and its impact on the accuracy of the stator flux linkages and hence on the electromagnetic torque is high when the stator voltages are small and of comparable magnitude to the resistive voltage drops. This is the case at low speeds; hence, dynamic operation at such speeds is very poor under this scheme.

Example 8.1

Find the errors in the stator flux-linkages magnitude and electromagnetic torque computations when the stator resistance has risen from its nominal value by 50%. The motor parameters are as follows:

$$R_s = 0.277 \, \Omega; R_r = 0.183 \, \Omega; L_m = 0.0538 \, \text{H}; L_s = 0.0553 \, \text{H}; L_r = 0.05606 \, \text{H}; P = 4$$

Operating points: (i) $\omega_s = 16.932 \, \text{rad/sec}$; $\omega_{sl} = 6.932 \, \text{rad/sec}$; $V = 17.657 \, \text{(line to line) rms}$
 (ii) $\omega_s = 306.932 \, \text{rad/sec}$; $\omega_{sl} = 6.932 \, \text{rad/sec}$; $V = 172.33 \, \text{(line to line) rms}$

Solution For ease of computation, the synchronous reference frame is employed.

Operating point (i): The synchronous-frame q and d axes voltages are

$$V_{qs}^e = \frac{\sqrt{2}}{\sqrt{3}} V = 14.416 \, \text{v}; V_{ds}^e = 0$$

The currents are found as,

$$\begin{bmatrix} I_{qs}^e \\ I_{ds}^e \\ I_{qr}^e \\ I_{dr}^e \end{bmatrix} = \begin{bmatrix} R_{sl} & L_s \omega_s & 0 & L_m \omega_s \\ -L_s \omega_s & R_{sl} & -L_m \omega_s & 0 \\ 0 & L_m \omega_{sl} & R_r & L_r \omega_{sl} \\ -L_m \omega_{sl} & 0 & -L_r \omega_{sl} & R_r \end{bmatrix}^{-1} \begin{bmatrix} V_{qs}^e \\ V_{ds}^e \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 17.641 \\ 5.176 \\ -15.722 \\ 2.459 \end{bmatrix}$$

where $R_{sl} = 1.5R_s$.

These machine currents are available through current transducers to the flux and torque processor. The stator flux linkages are then calculated, with the nominal value of the stator resistance in the calculator circuit as

$$\lambda_{qs}^e(R_s) = -\frac{V_{ds}^e - R_s I_{ds}^e}{\omega_s}; \quad \lambda_{ds}^e(R_s) = \frac{V_{qs}^e - R_s I_{qs}^e}{\omega_s}$$

$$\lambda_s(R_s) = \sqrt{\{\lambda_{qs}^e(R_s)\}^2 + \{\lambda_{ds}^e(R_s)\}^2}$$

from which, upon substitution of the stator currents and voltages for stator resistances of R_s and R_{s1} , the stator flux linkages are obtained as

$$\lambda_{qs}^e(R_s) = 0.085 \, \text{Wb-Turn}; \lambda_{ds}^e(R_s) = 0.563 \, \text{Wb-Turn}; \lambda_s(R_s) = 0.569 \, \text{Wb-Turn}$$

$$\lambda_{qs}^e(R_{s1}) = 0.127 \, \text{Wb-Turn}; \lambda_{ds}^e(R_{s1}) = 0.419 \, \text{Wb-Turn}; \lambda_s(R_{s1}) = 0.437 \, \text{Wb-Turn}$$

$$T_e(R_s) = \frac{3}{2} \frac{P}{2} \{\lambda_{ds}^e(R_s) I_{qs}^e - \lambda_{qs}^e(R_s) I_{ds}^e\}$$

from which the torque is calculated as

$$T_e(R_s) = 28.473 \text{ N}\cdot\text{m}; \text{ and } T_e(R_{s1}) = 20.179 \text{ N}\cdot\text{m}$$

The errors in the flux and torque are

$$\partial\lambda_s = \lambda_s(R_{s1}) - \lambda_s(R_s) = -0.132 \text{ Wb-Turn}$$

$$\partial T_e = T_e(R_{s1}) - T_e(R_s) = -8.249 \text{ N}\cdot\text{m}$$

These errors are considerable; the flux error is approximately 25% and torque error is 41%, compared to rated values. Consider the second operating point.

Operating point (ii): The currents at R_{s1} are found, similarly to the above procedure.

$$I_{qs}^e = 15.67 \text{ A}; I_{ds}^e = 9.614 \text{ A}$$

The flux linkages are

$$\lambda_{qs}^e(R_s) = 0.009 \text{ Wb-Turn}; \lambda_{ds}^e(R_s) = 0.444 \text{ Wb-Turn}; \lambda_s(R_s) = 0.444 \text{ Wb-Turn}$$

$$\lambda_{qs}^e(R_{s1}) = 0.013 \text{ Wb-Turn}; \lambda_{ds}^e(R_{s1}) = 0.437 \text{ Wb-Turn}; \lambda_s(R_{s1}) = 0.437 \text{ Wb-Turn}$$

The torques are

$$T_e(R_s) = 20.636 \text{ N}\cdot\text{m}; T_e(R_{s1}) = 20.178 \text{ N}\cdot\text{m}$$

The errors in the stator flux linkages and torque are

$$\partial\lambda_s = \lambda_s(R_{s1}) - \lambda_s(R_s) = -0.007 \text{ Wb-Turn}$$

$$\partial T_e = T_e(R_{s1}) - T_e(R_s) = -0.458 \text{ N}\cdot\text{m}$$

These errors are very small; they constitute less than 2% of rated values.

8.3.2.2 Case (ii): Induced EMF from flux-sensing coils or hall sensors.

Two sets of sensing coils can be placed in stator slots having 90 electrical degrees displacement, and one set can be placed on the mmf axis of one phase, say phase a . These coils can be concentric, making the layout easier. The sensing coils are isolated from the power circuit, to help in tying the outputs of those coils to the logic level-control circuits directly. Two Hall sensors can also be placed very similarly to the sensing coils; Hall sensors reflect the rate of change of the stator flux linkages. Let the induced emfs of the q and d axes be e_{qs} and e_{ds} , respectively. The stator q and d axes flux linkages, λ_{qs} and λ_{ds} , respectively, are defined as

$$\lambda_{qs} = L_s i_{qs} + L_m i_{qr} \quad (8.14)$$

$$\lambda_{ds} = L_s i_{ds} + L_m i_{dr}$$

In terms of induced emfs, they are

$$\begin{aligned} \lambda_{qs} &= \int e_{qs} dt \\ \lambda_{ds} &= \int e_{ds} dt \end{aligned} \quad (8.15)$$

from which the rotor q and d axes currents are derived as,

$$i_{qr} = \frac{\lambda_{qs}}{L_m} - \frac{L_s}{L_m} i_{qs} = \frac{1}{L_m} \{ \int e_{qs} dt - L_s i_{qs} \} \quad (8.16)$$

$$i_{dr} = \frac{\lambda_{ds}}{L_m} - \frac{L_s}{L_m} i_{ds} = \frac{1}{L_m} \{ \int e_{qs} dt - L_s i_{qs} \} \quad (8.17)$$

From the rotor q and d axes currents and stator flux linkages, the rotor flux linkages and torque are obtained as

$$\begin{aligned} \lambda_{qr} &= L_r i_{qr} + L_m i_{qs} \\ \lambda_{dr} &= L_r i_{dr} + L_m i_{ds} \\ \lambda_r &= \sqrt{(\lambda_{qr})^2 + (\lambda_{dr})^2} \angle \theta_f \\ \theta_f &= \tan^{-1} \left(\frac{\lambda_{qr}}{\lambda_{dr}} \right) \\ T_e &= \frac{3}{2} \frac{P}{2} (\lambda_{ds} i_{qs} - \lambda_{qs} i_{ds}) \end{aligned} \quad (8.18)$$

Note that the torque equation does not involve machine parameters for its calculation. The rotor flux linkages and the field angle are dependent on machine parameters: L_m , L_s , and L_r . When there is a change (due to saturation) in these parameters, they will introduce an error in the computation of λ_r and θ_f . In particular, the error in the computation of field angle will generate significant errors in the vector control; that is the most crucial information for control. The flux and torque processor realized with terminal or sensing coils or Hall sensors have some merits and demerits. These are discussed in the following.

The merits of these forms of measuring and computing the rotor flux linkages, its position, and the electromagnetic torque are as follows.

- (i) The sensing schemes use only electronic transducers and do not use any with moving or rotating parts, such as synchros or optical encoders. The absence of moving parts in the transducers makes the reliability of these schemes more robust than from employing mechanical/optical transducers.
- (ii) Further, the costly process of mechanical mounting and the loss of valuable space and volume inside or outside the machine enclosure for the rotating sensor parts are avoided. The compactness afforded increases the overall power density of the motor drive system. A saving in labor and parts makes these sensing schemes very attractive at the low-cost, low-power end of industrial applications.

Some disadvantages are as follows:

- (i) At zero stator frequency, there is no induced emf in all the measurement schemes. The result is that neither flux linkages nor their positions are available

- for vector control. Therefore, torque production at zero speed is not precisely controllable, thus making it unsuitable for positioning applications such as servos.
- (ii) The same problem in a different form appears at low speeds. At these speeds, the induced-emf signals can be so small that signal-processing circuits cannot use them; they are comparable to quantization errors in digital circuits and will be affected by drift in the analog circuits. This, combined with factor (i) makes the direct vector-control drive unsuitable for precise positioning and low-speed operation.
 - (iii) The installation of sensing windings or of Hall-effect sensors, even though it is an inexpensive production process, adds to the number of wires coming out of the machine frame. This is not acceptable in high-volume applications, such as HVAC, because of the cost involved in the hermetic sealing. Extra wires are also not desirable in high-reliability applications, such as defense actuators and nuclear-plant pump drives.
 - (iv) In the case of using voltage and current transducers, the filtering required to obtain the fundamental at high frequencies will produce a large phase shift and inaccuracy in the computation of field angle, which will deteriorate the decoupling of flux and torque controls. It is relevant to observe here that voltage sensing with galvanic isolation can be easily and cost-effectively realized from the logic-level switching signals of the inverter, the phase currents, and an indication of the dc link voltage magnitude. The interested reader is referred to [25].

8.3.3 Implementation with Six-Step Current Source

In the above development, the form of current-source inverter was not discussed. It could be a six-step inverter or an inverter with sinusoidal current outputs. The six-step current-source inverter can be realized either with an ASCI or with a PWM inverter with inner current loops, discussed in Chapter 7. The sinusoidal inverter is realized from a PWM inverter with inner current loops. The switching of the inverter can be done with different modulation techniques. This section considers the six-step current-source inverter for direct vector-control implementation. Some drives use an ASCI implementation for its advantages in cost and reliability.

The simulation results of direct vector control with a six-step current-source inverter is shown in Figures 8.4(i) and (ii). Ideal current source is assumed with negligible rise and fall times. One-p.u. speed is commanded from standstill, which produces the maximum torque command. The rotor flux linkages are at 1 p.u. when the speed command is applied. The torque command is limited to ± 2 p.u. In a six-step current-source inverter, a high amount of sixth-harmonic torque pulsation is present. A feedback of such a torque results in larger oscillations in torque in a direct vector-controlled induction motor. To avoid such a problem, either the feedback control of torque and flux linkages is deleted or the filtered torque signal is used for feedback control. These two cases are shown in Figures 8.4(i) and (ii), respectively. In both the control schemes, the key results are similar, except in the case of flux linkages experiencing a deviation in the open-loop case. Also, because of the torque

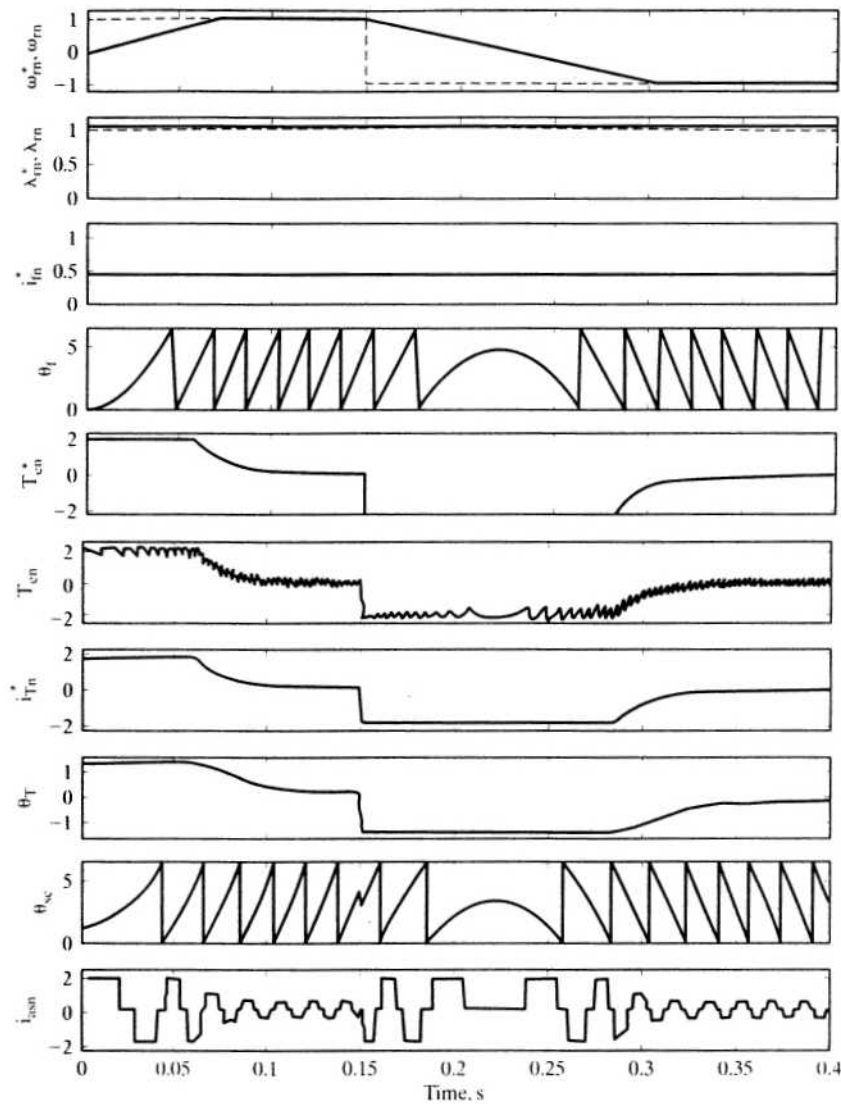


Figure 8.4(i) Dynamic responses of a speed-controlled six-step current-source-based direct vector-controlled induction motor drive (no T_e , λ_r feedback)

feedback, there is additional switching introduced in the phase currents. The common performance features are summarized in the following.

The electromagnetic torque rises instantaneously, and its average reaches the command value. A small swing in rotor flux linkages is seen, due to the fact that the stator-current phasor can assume only seven distinct states and hence is not smooth. As the speed error is decreased, the torque command and, hence, the magnitudes of the stator currents decrease, as is clearly seen in the figure.

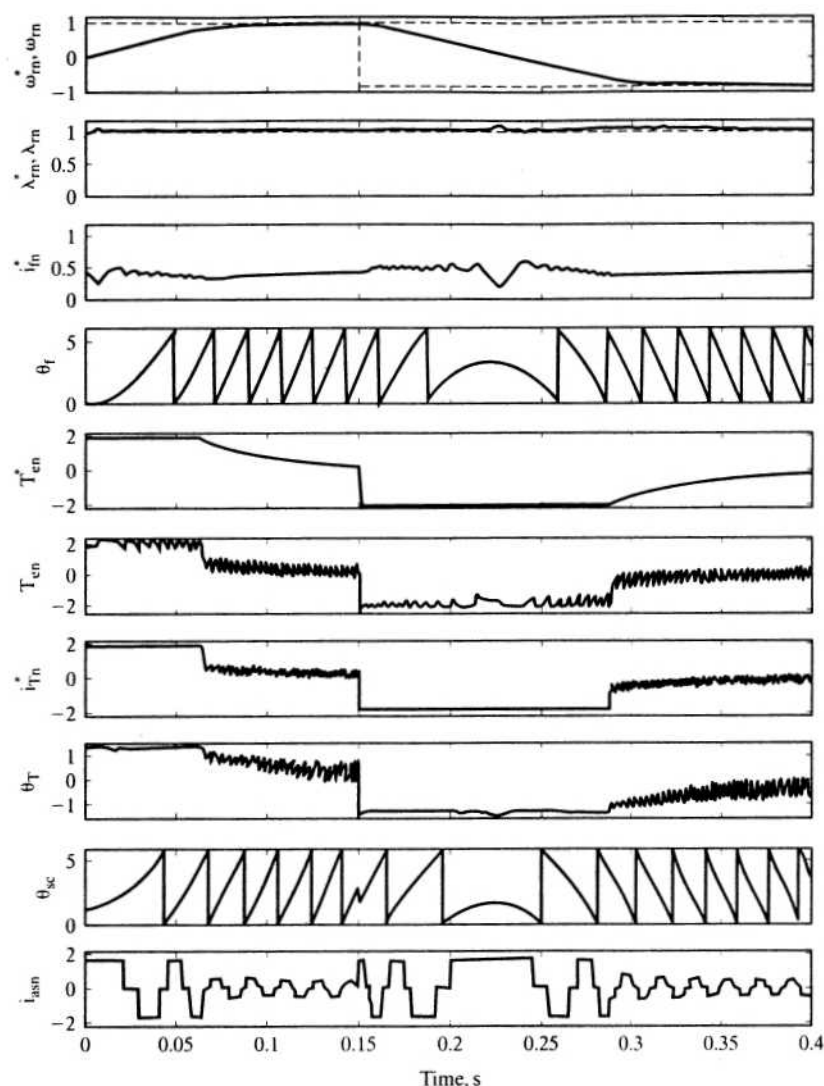


Figure 8.4(ii) Dynamic responses of a speed-controlled six-step current-source-based direct vector-controlled induction motor drive (with T_e , λ_r feedback loops included)

Figure 8.5 shows the simulation results of a sinusoidal PWM inverter-controlled induction motor torque drive. A step command of one-p.u. bidirectional load torque is applied. The torque-producing current rises in a very short time to match the load torque, and the response is almost instantaneous. As the speed is allowed to vary freely, the stator frequency follows the speed as seen from the stator current. The rotor flux linkages are steady and are unperturbed by the torque changes.

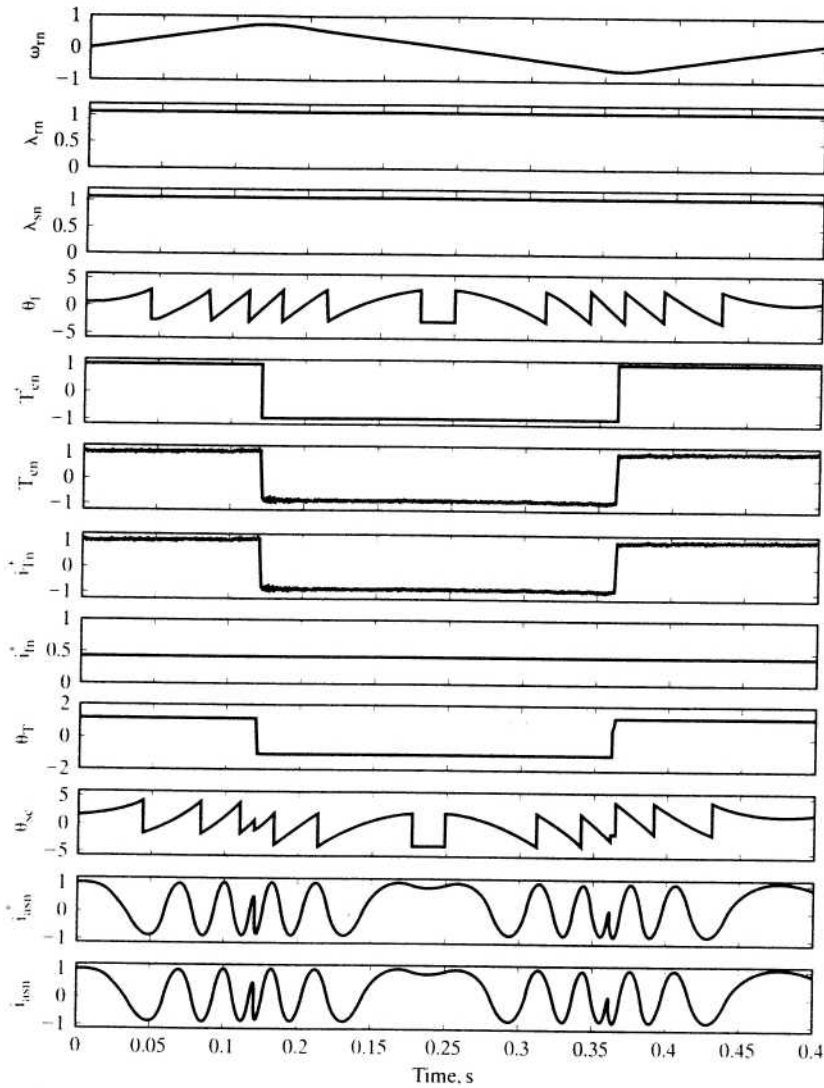


Figure 8.5 Output characteristics for a torque-controlled current-source direct vector-control scheme

8.3.4 Implementation with Voltage Source

Figure 8.6 shows the vector-control implementation with a voltage-source inverter. The current commands i_{qs}^* and i_{ds}^* are generated, very much as with the vector control with current-source inverter. The dq current errors set the required stator voltage commands v_{qs}^* and v_{ds}^* . By using the transformation matrix, the two axis voltage commands are converted into three-phase voltage commands, which will

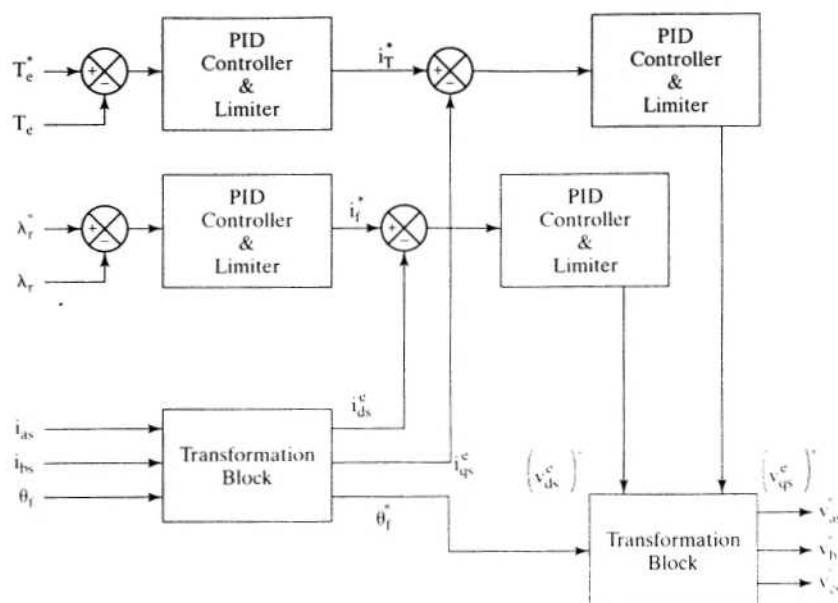


Figure 8.6 Functional block diagram of a voltage-source direct vector controller

be amplified by the inverter using any one of the switching techniques. The simulation results for a speed-controlled drive using PWM voltage source are shown in Figure 8.7. An 0.75-p.u. bidirectional speed command is given to the drive system when the machine has rated rotor flux linkages. During acceleration and deceleration, the torque is maintained at 2 p.u. and -2 p.u., respectively. Note the smooth acceleration and deceleration speed profiles. The zero-speed crossover is affected, with no oscillations in current, torque, flux, and speed, by means of the precise control of rotor flux linkages.

8.3.5 Direct Vector (Self) Control in Stator Reference Frames with Space-Vector Modulation

A novel way of implementing the direct vector-controlled induction motor drive with voltage-source inverter is presented in this section from reference [46]. This method uses feedback control of torque and stator flux, which are computed from the measured stator voltages and currents. As the method does not use a position or speed sensor to control the machine and uses its own electrical output currents and resulting terminal voltages, this is also referred to as a direct self-control scheme. The method uses a stator reference model of the induction motor for its implementation, thereby avoiding the trigonometric operations in the coordinate transformations of the synchronous reference frames. This is one of the key advantages of the control scheme. The scheme uses stator flux-linkages control. That enables the flux-weakening operation of the motor to be straightforward compared to rotor flux-weakening, in that the stator flux is directly proportional to the induced emf, whereas the rotor flux does not have the same relationship. This reduces the depen-

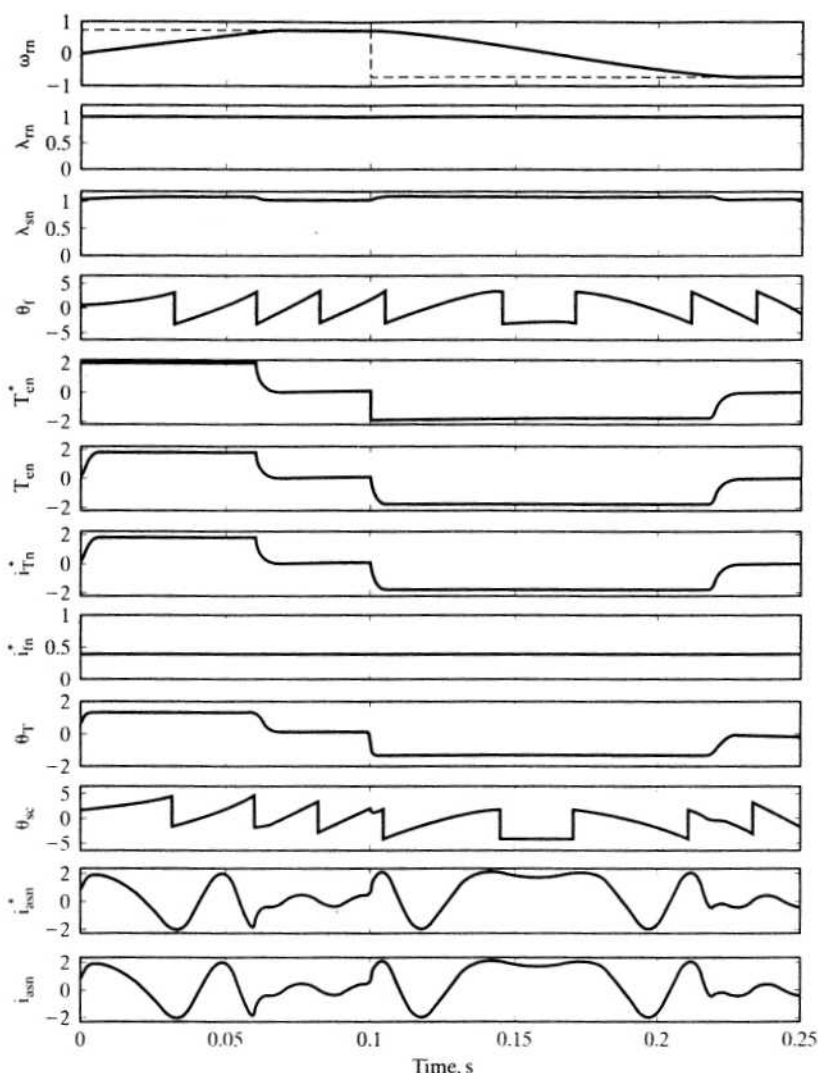


Figure 8.7 Dynamic performance of a speed-controlled voltage-source direct vector-control scheme

dence of the scheme on many motor parameters, thus making it a robust scheme in the flux-weakening region. The scheme depends only on stator resistance and on no other parameters. For its flux and torque control, this control scheme requires the position of the flux phasor, which is difficult to obtain at low and zero speeds from measured voltages and currents by using a torque processor. At such low speeds, these signals are very small, making scaling and accuracy problematic; also, the stator-resistance variations introduce significant errors in the flux-phasor computation. Invariably the low-speed operation suffers with this scheme without stator-resistance compensation.

The implementation of the scheme requires flux linkages and torque computations, plus generation of switching states through a feedback control of the torque and flux directly without inner current loops.

The stator q and d axes flux linkages are

$$\lambda_{qs} = \int (v_{qs} - R_s i_{qs}) dt \quad (8.19)$$

$$\lambda_{ds} = \int (v_{ds} - R_s i_{ds}) dt \quad (8.20)$$

where the direct and quadrature axis components are obtained from the abc variables by using the transformation,

$$i_{qs} = i_{as} \quad (8.21)$$

$$i_{ds} = \frac{1}{\sqrt{3}} (i_{cs} - i_{bs}) \quad (8.22)$$

This transformation is applicable for voltages and flux linkages as well. To obtain uniformly rotating stator flux, note that the motor voltages have to be varied uniformly without steps too. This imposes a requirement of continuously variable stator voltages with infinite steps, which is not usually met by the inverter because it has only finite switching states.

Switching States of the Inverter Consider the inverter shown in Figure 8.8. The terminal voltage a with respect to negative of the dc supply is considered, and V_a is determined by a set of switches, S_a , consisting of T_1 and T_4 as shown in the Table 8.1. When the switching devices T_1 and T_4 and their antiparallel diodes are off, V_a is indeterminate. Such a situation is not encountered in practice and, hence, has not been considered. The switching of S_b and S_c sets for line b and c can be similarly derived. The total number of switching states possible with S_a , S_b , and S_c is eight; they are elaborated in Table 8.2 by using the following relationships:

$$\left. \begin{aligned} v_{ab} &= v_a - v_b \\ v_{bc} &= v_b - v_c \\ v_{ca} &= v_c - v_a \end{aligned} \right\} \quad (8.23)$$

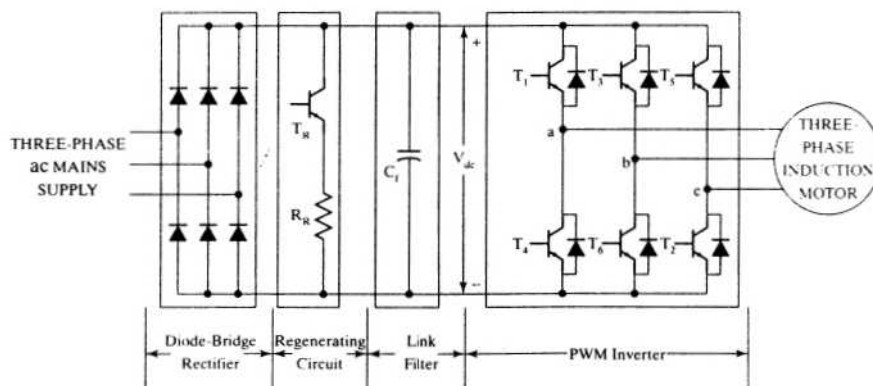


Figure 8.8 Power-circuit configuration of the induction motor drive

TABLE 8.1 Switching states of inverter phase leg *a*

T_1	T_4	S_a	V_a
on	off	1	V_{dc}
off	on	0	0

TABLE 8.2 Inverter switching states and machine voltages

States	S_a	S_b	S_c	V_a	V_b	V_c	V_{ab}	V_{bc}	V_{ca}	V_{as}	V_{bs}	V_{cs}	V_{qs}	V_{ds}
I	1	0	0	V_{dc}	0	0	V_{dc}	0	$-V_{dc}$	$\frac{2}{3}V_{dc}$	$-\frac{1}{3}V_{dc}$	$-\frac{1}{3}V_{dc}$	$\frac{2}{3}V_{dc}$	0
II	1	0	1	V_{dc}	0	V_{dc}	V_{dc}	$-V_{dc}$	0	$\frac{1}{3}V_{dc}$	$-\frac{2}{3}V_{dc}$	$\frac{1}{3}V_{dc}$	$\frac{1}{3}V_{dc}$	$\frac{V_{dc}}{\sqrt{3}}$
III	0	0	1	0	0	V_{dc}	0	$-V_{dc}$	V_{dc}	$-\frac{1}{3}V_{dc}$	$-\frac{1}{3}V_{dc}$	$\frac{2}{3}V_{dc}$	$-\frac{1}{3}V_{dc}$	$\frac{V_{dc}}{\sqrt{3}}$
IV	0	1	1	0	V_{dc}	V_{dc}	$-V_{dc}$	0	V_{dc}	$-\frac{2}{3}V_{dc}$	$\frac{1}{3}V_{dc}$	$\frac{1}{3}V_{dc}$	$-\frac{2}{3}V_{dc}$	0
V	0	1	0	0	V_{dc}	0	$-V_{dc}$	V_{dc}	0	$-\frac{1}{3}V_{dc}$	$\frac{2}{3}V_{dc}$	$-\frac{1}{3}V_{dc}$	$-\frac{1}{3}V_{dc}$	$-\frac{V_{dc}}{\sqrt{3}}$
VI	1	1	0	V_{dc}	V_{dc}	0	0	V_{dc}	$-V_{dc}$	$\frac{1}{3}V_{dc}$	$\frac{1}{3}V_{dc}$	$-\frac{2}{3}V_{dc}$	$\frac{1}{3}V_{dc}$	$-\frac{V_{dc}}{\sqrt{3}}$
VII	0	0	0	0	0	0	0	0	0	0	0	0	0	0
VIII	1	1	1	V_{dc}	V_{dc}	V_{dc}	0	0	0	0	0	0	0	0

and machine phase voltages for a balanced system are

$$\left. \begin{aligned} v_{as} &= \frac{(v_{ab} - v_{ca})}{3} \\ v_{bs} &= \frac{(v_{bc} - v_{ab})}{3} \\ v_{cs} &= \frac{(v_{ca} - v_{bc})}{3} \end{aligned} \right\} \quad (8.24)$$

and q and d axes voltages are given by

$$\left. \begin{aligned} v_{qs} &= v_{as} \\ v_{ds} &= \frac{1}{\sqrt{3}}(v_{cs} - v_{bs}) = \frac{1}{\sqrt{3}}v_{cb} \end{aligned} \right\} \quad (8.25)$$

The stator q and d voltages for each state are shown in Figure 8.9. The limited states of the inverter create distinct discrete movement of the stator-voltage phasor, v_s , consisting of the resultant of v_{qs} and v_{ds} . An almost continuous and uniform flux phasor is feasible with these discrete voltage states, one due to their integration over time as seen from equations (8.19) and (8.20).

For control of the voltage phasor both in its magnitude and phase, the requested voltage vector's phase and magnitude are sampled, say once every

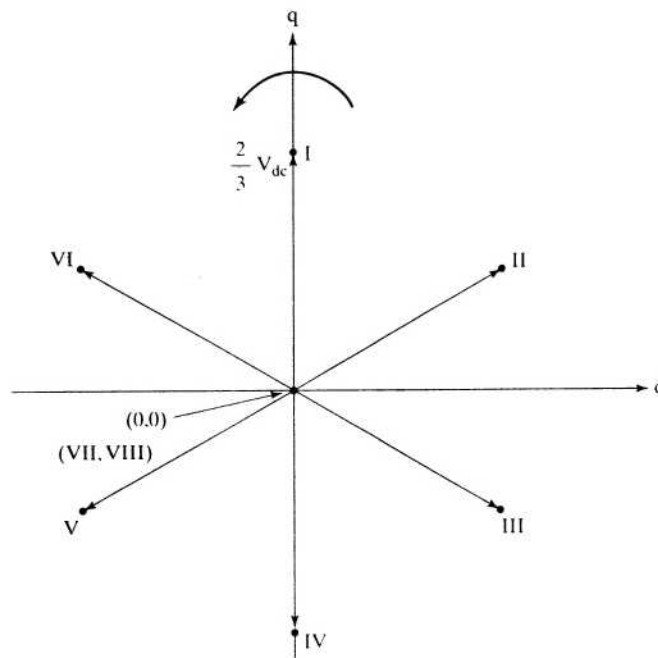


Figure 8.9 The inverter output voltages corresponding to switching states

switching period. The phase of the requested voltage vector identifies the nearest two nonzero voltage vectors. The requested voltage vector can be synthesized by using fractions of the two nearest voltage vectors, which amounts to applying these two vectors, one at a time, for a fraction of the switching period. The nearest zero voltage vector to the two voltage vectors is applied for the remaining switching period. The duty cycle for each of the voltage vectors is determined by the phasor projection of the requested voltage vector onto the two nearest voltage vectors. This method of controlling the input voltages to the machine through a synthesis of voltage phasor rather than the individual line-to-line voltages has the advantages of: (i) not using pulse-width modulation carrier-frequency signals, (ii) higher fundamental voltages compared to sine-triangle PWM, based controllers, (iii) given the switching frequency, the switching losses are minimized, and (iv) lower voltage and current ripples. This method of switching the inverter is known as space vector modulation (SVM) and many variations are available in literature.

Flux control A uniform rotating stator flux is desirable, and it occupies one of the sextants (in the phasor diagram shown in Figure 8.10) at any time. The stator-flux phasor has a magnitude of λ_s , with an instantaneous position of θ_s . The corresponding d and q axes components are λ_{ds} and λ_{qs} , respectively. Assuming that a feedback of stator flux is available, its place in the sextant is identified from its position. Then the influencing voltage phasor is identified by giving a 90° phase shift. For example, if the stator-flux phasor is in sextant $\langle 2 \rangle$, the right influencing

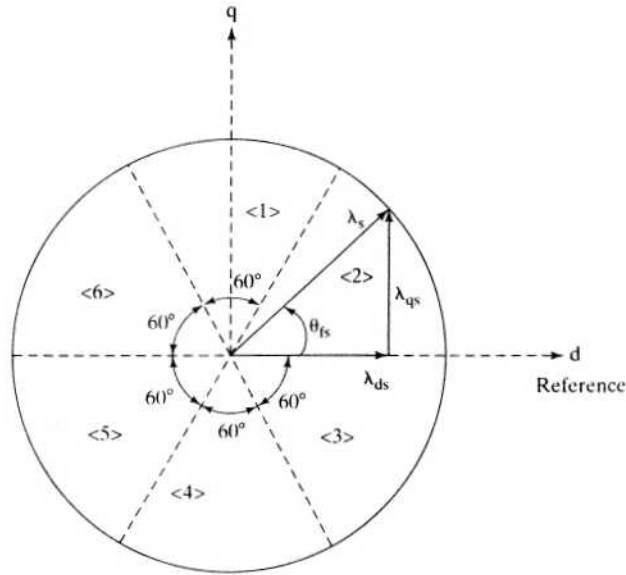


Figure 8.10 Division of sextants for stator flux-linkages identification

voltage phasor has to be either VI or I. Voltage phasor I is $90^\circ - \theta_{fs}$ and VI is $150^\circ - \theta_{fs}$ from the flux phasor. One of these two sets increases λ_s , the other decreases λ_s . This is found from the following explanation obtained from Figure 8.11.

Consider the effect of switching voltage phasor set I, v_{I1} , and phasor set VI, v_{VI1} . As seen from the phasor diagram, in the case of set I, the flux phasor increases in magnitude from λ_s to λ_{sI} ; in the case of set VI, it decreases to λ_{sVI} . This implies that the closer voltage-phasor set increases the flux and the farther voltage-phasor set decreases the flux, but note that both of them advance the flux phasor in position. Similarly for all other sextants, the switching logic is developed. A flux error, λ_{er} , thus determines which voltage phasor has to be called, and this flux error is converted to a digital signal with a window comparator with a hysteresis of $\delta\lambda_s$ (let it be S_λ). The switching logic to realize S_λ from λ_{er} is given in the following:

Condition	S_λ
$\lambda_{er} > \lambda_s$	1
$\lambda_{er} \leq \lambda_s$	0

Over and above the flux control, the control of electromagnetic torque is required for a high-performance drive. It is achieved as follows.

Torque control Torque control is exercised by comparison of the command torque to the torque measured from the stator flux linkages and stator currents as

$$T_e = \frac{3}{2} \cdot \frac{P}{2} (\lambda_{ds} i_{qs} - \lambda_{qs} i_{ds}) \quad (8.25)$$

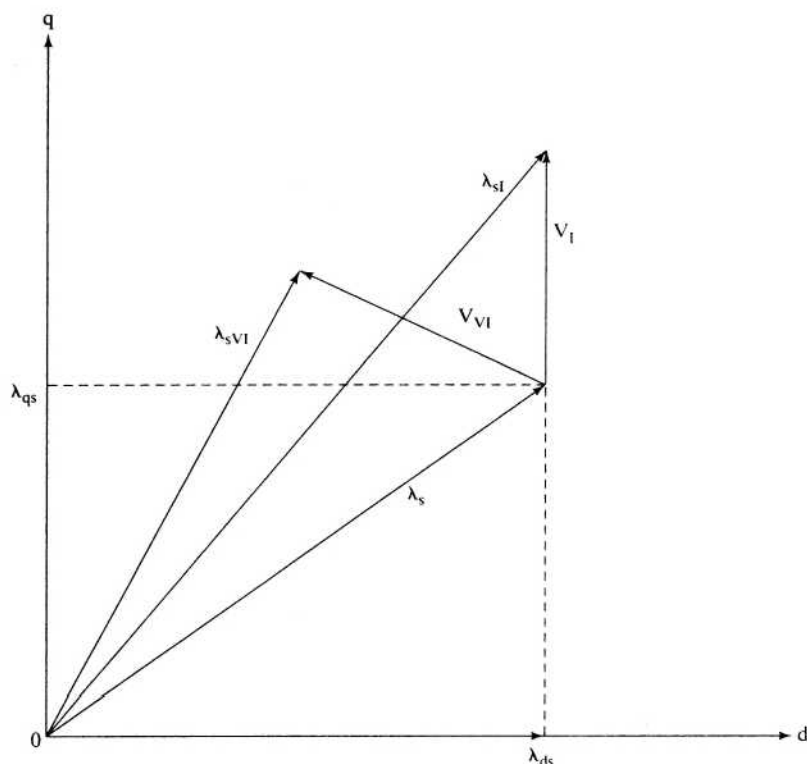


Figure 8.11 Effect of switching v_l and v_{vl} on stator-flux phasor

The error torque is processed through a window comparator to produce digital outputs, S_T , as follows:

Condition	S_T
$(T_e^* - T_e) > \delta T_e$	1
$-\delta T_e < (T_e^* - T_e) < \delta T_e$	0
$(T_e^* - T_e) < -\delta T_e$	-1

where δT_e is the torque window acceptable over the commanded torque. When the error exceeds δT_e , it is time to increase the torque, denoting it with a +1 signal. If the torque error is between positive and negative torque windows, then the voltage phasor could be at zero state. If the torque error is below $-\delta T_e$, it amounts to calling for regeneration, signified by -1 logic signal.

Interpretation of S_T is as follows: when it is 1 amounts to increasing the voltage phasor, 0 means to keep it at zero, -1 requires retarding the voltage phasor behind the flux phasor to provide regeneration. Combining the flux error output S_λ , the torque error output S_T , and the sextant of the flux phasor S_θ , a switching table can be realized to obtain the switching states of the inverter; it is given in Table 8.3. The algorithm for S_θ is shown in Table 8.4.

TABLE 8.3 Switching states for possible S_λ , S_T , and S_θ

S_λ	S_T	S_θ					
		<1>	<2>	<3>	<4>	<5>	<6>
1	1	VI (1,1,0)	I (1,0,0)	II (1,0,1)	III (0,0,1)	IV (0,1,1)	V (0,1,0)
1	0	VIII (1,1,1)	VII (0,0,0)	VIII (1,1,1)	VII (0,0,0)	VIII (1,1,1)	VII (0,0,0)
1	-1	II (1,0,1)	III (0,0,1)	IV (0,1,1)	V (0,1,0)	VI (1,1,0)	I (1,0,0)
0	1	V (0,1,0)	VI (1,1,0)	I (1,0,0)	II (1,0,1)	III (0,0,1)	IV (0,1,1)
0	0	VII (0,0,0)	VIII (1,1,1)	VII (0,0,0)	VIII (1,1,1)	VII (0,0,0)	VIII (1,1,1)
0	-1	III (0,0,1)	IV (0,1,1)	V (0,1,0)	VI (1,1,0)	I (1,0,0)	II (1,0,1)

TABLE 8.4 Flux-phasor sextant logic (S_θ)

θ_{fs}	Sextant
$0 \leq \theta_{fs} \leq \pi/3$	<2>
$-\pi/3 \leq \theta_{fs} \leq 0$	<3>
$-2\pi/3 \leq \theta_{fs} \leq -\pi/3$	<4>
$-\pi \leq \theta_{fs} \leq -2\pi/3$	<5>
$2\pi/3 \leq \theta_{fs} \leq \pi$	<6>
$\pi/3 \leq \theta_{fs} \leq 2\pi/3$	<1>

Consider the first column corresponding to $S_\theta = <1>$. The switching states of the inverter are indicated in parentheses; they correspond to S_a , S_b , and S_c . The flux error signal indicates 1, which means the flux is less than its request value and therefore the flux phasor has to be increased. At the same time, torque error is positive, asking for an increase. Merging these two with the position of the flux phasor in <1>, the voltage phasor I and VI satisfy the requirements only if the flux is within the first 30° of the sextant <1>. In the second 30°, note that the voltage phasor I will increase the flux-phasor magnitude but will retard it in phase. This will result in a reduction of the stator frequency and reversal of the direction of torque. The control requires the advancement of the flux phasor in the same direction (i.e., counterclockwise in this discussion); that could be satisfied only by voltage phasor VI in this 30°. Voltage phasor VI is the only one satisfying the uniform requirements throughout the sextant <1>, so the voltage phasor VI is chosen for S_T and S_λ to be equal to +1 with the flux phasor in sextant <1>. When the torque error is zero, the only logical choice is to apply zero line voltages; because the previous state had two +1 states, it is easy to achieve zero line voltages by choosing the switching state VIII with all ones. If the torque error becomes negative with $S_T = -1$, the machine has to regenerate, with a simultaneous increase in flux phasor due to $S_\lambda = 1$; hence, the voltage phasor is retarded close to the flux phasor, and, hence, switching state II(1, 0, 1) is selected.

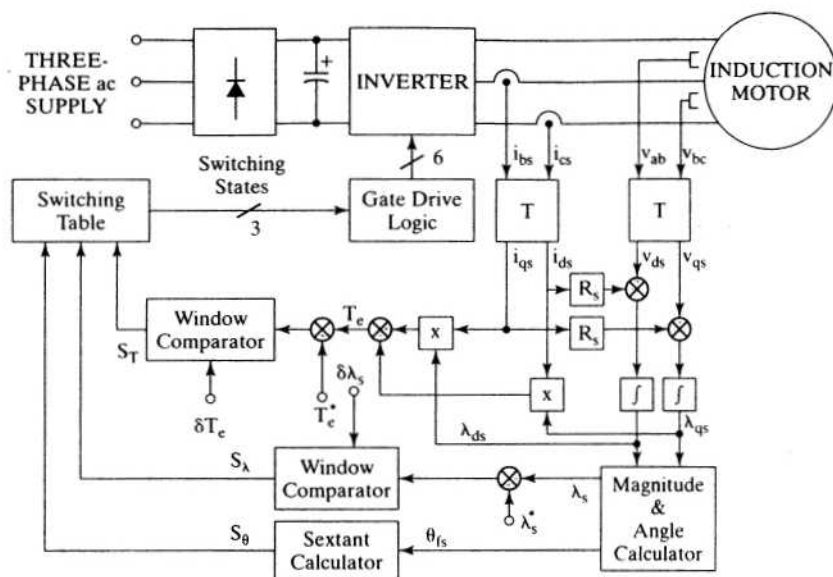


Figure 8.12 Block-diagram schematic of the direct torque (self) induction motor drive

If $S_\lambda = 0$ (i.e., the flux phasor has exceeded its request by the hysteresis window amount, $\delta\lambda_s$), then it has to be decreased to match its request value by choosing a voltage phasor away from flux phasor, i.e., V. This accelerates the flux phasor and increases the slip speed, resulting also in an increase of electromagnetic torque, thus satisfying $S_T = 1$ demand. When $S_T = 0$, reach zero line-voltage states by going to VII, because V contained two zero states in it. If $S_T = -1$, then regenerate, increasing the negative torque but decreasing the flux phasor; this is achieved by retarding the voltage phasor behind the flux phasor, but far away from it, and hence III is chosen. Note that it has two zeros in it, and, therefore, transition from VII to III requires a change of only one switch signal.

Implementation The drive scheme is realized as shown in Figure 8.12. Further, to reduce the voltage transducers, the information of line and phase voltages could be obtained from a single dc-link voltage transducer and the gate drive signals. Similarly, the phase currents can be reconstructed from a single dc-link current transducer and the gate drive signals of the inverter. In all, only two transducers are required, both of which are electrical; it does not require moving parts, thus making this control scheme robust and reliable. Further, the cost of the drive-system control is very low compared to that of position-sensor-based vector-controlled induction motor drives.

Performance The dynamic performance of this drive scheme is shown for both the torque- and the speed-controlled drive in Figures 8.13(a) and (b), respectively. The motor details are given in Example 5.4. The hysteresis windows for the torque and flux linkages are 0.05 p.u. and 0.001 p.u., respectively. The proportional and integral gains of the speed controller are 50 and 0.05, respectively. The base values of torque, speed, flux linkage, current, and voltage are 20.3 N·m, 377 rad/sec, 0.4213 Wb-turn,

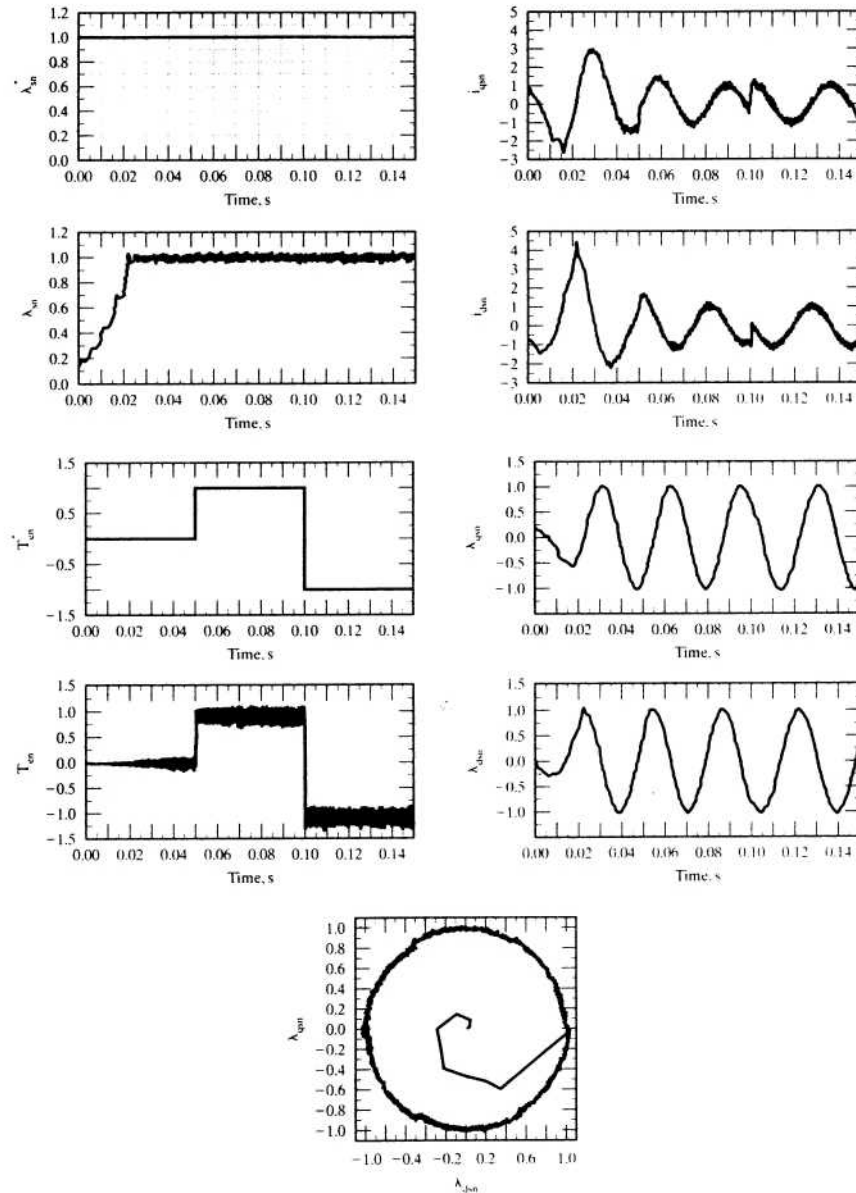


Figure 8.13(a) Dynamic performance of the torque-controlled drive system in normalized units

18.94 A, and 200 V, respectively. The dc-link voltage is set at 285 V. The rotor is free of load for both torque- and speed-controlled operation.

For the torque drive, note that the machine starts from standstill and, hence, takes a time of 25 ms to establish the stator flux linkages with high currents of 4.5 p.u. This is unusual and is not attempted in practice. The stator flux phasor

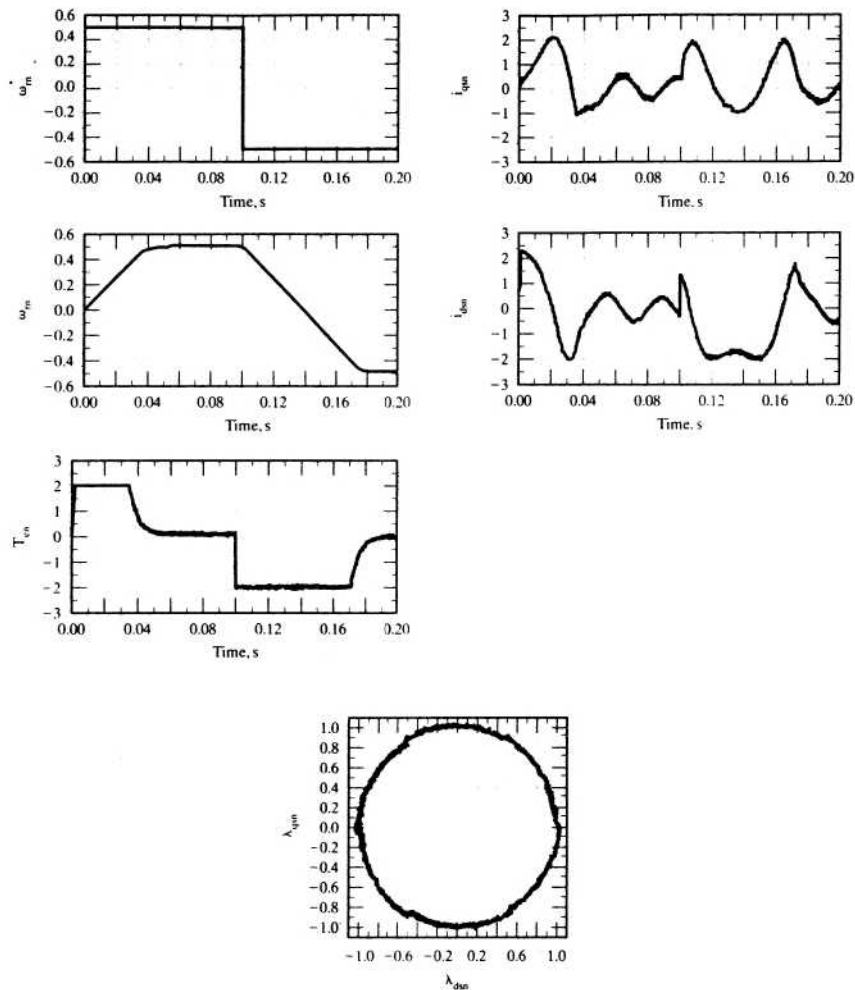


Figure 8.13(b) Dynamic performance of the speed-controlled drive system in normalized units

becomes uniform, as seen from its plot. The torque response is almost instantaneous, with considerable switching frequency ripples on it. The ripple torques have no effect on the speed, as shown from the speed-controlled drive.

The speed command is a step of 0.5 p.u. at starting and a step of 0.5 to -0.5 p.u. at 0.1 s. It is assumed that the machine has rated flux linkages at the time of application of the speed command. The electromagnetic torque is limited to ± 2 p.u., which provides a faster acceleration, as is seen from the Figure 8.13(b). The rotor speed takes a smooth ramp profile with no oscillations, while the current is limited to two p.u. The locus of the stator flux-linkages phasor is almost a uniform circle, even during large speed changes and hence torque commands, thus showing the complete decoupling of the flux- from the torque-producing channels in the drive system.

Example 8.2

It is noticed that stator frequency is not a distinct input in the direct self-control scheme. How can that be extracted for possible use in speed-sensorless control application?

Solution The synchronous speed is the speed of the stator flux-linkages phasor. It may be recalled that the synchronous frequency is equal to the stator frequency. It is obtained as the derivative of the stator flux-linkages-phasor position. As the stator flux linkages and its d and q components are available, this task is made easier in the control scheme. The synchronous speed, then, is derived as

$$\begin{aligned}\omega_s &= \frac{d\theta_{fs}}{dt} = \frac{d}{dt} \left\{ \tan^{-1} \left(\frac{\lambda_{qs}}{\lambda_{ds}} \right) \right\} \\ &= \frac{\dot{\lambda}_{qs}\lambda_{ds} - \dot{\lambda}_{ds}\lambda_{qs}}{\lambda_{ds}^2 + \lambda_{qs}^2} \\ &= \frac{(v_{qs} - R_s i_{qs})\lambda_{ds} - (v_{ds} - R_s i_{ds})\lambda_{qs}}{\lambda_s^2}\end{aligned}$$

Note that differentiation of the q and d axes stator flux linkages is avoided with measured stator voltages and computed resistive voltage drops with measured currents. The denominator is zero to start with, so caution must be exercised at start. After the initial start mode, the denominator, which is the stator flux-linkages phasor, is always nontrivial. As the stator voltages are pulsed, the stator frequency signal is fraught with noise and, therefore, filtering is required to enable its use in feedback control applications.

Parameter sensitivity The scheme is sensitive to stator resistance instrumented in the controller. Note that, in comparison to many other schemes, the parameter dependency of this scheme is restricted to only one parameter; other control schemes are dependent on many more parameters. The stator-resistance change has a wide variation (from 0.75 to 1.7 times its nominal value), due to a large extent to temperature variation and to a smaller degree to stator-frequency variation. It deteriorates the drive performance by introducing errors in the estimated flux linkage's magnitude and position and hence in the electromagnetic torque estimation, particularly at low speeds. Note that, at low speeds, the stator-resistance voltage drops constitute a significant portion of the applied voltages.

A few control schemes have been proposed to overcome this parameter sensitivity. A partial, operating-frequency-dependent hybrid flux estimator has been proposed for stator-resistance tuning; it has as problems convergence and slowness of response. Adjustment of the stator resistance in conformance to the difference between the flux current and its command has the problem of identifying the actual flux current. Finding stator resistance from the steady-state voltage equation has the shortcoming of using the direct axis flux linkage, which itself has been affected by stator-resistance variations. The use of stator-current phasor error with PI and fuzzy estimators demonstrates good performance in the tuning of the stator resistance. The fuzzy estimator seems to outperform the PI-based estimator, thereby necessitating the use of intelligent control techniques to synthesize the adaptation mechanism.

There is a possible instability problem, due to parameter sensitivity, that arises in the direct self-control scheme. The drive system becomes unstable if the instrumented

stator resistance in the controller is higher than the machine resistance. Such cases can arise in practice. Two possible scenarios follow:

- (i) The stator resistance can be lower than its controller-set-point nominal value in an externally housed drive system in colder climates; the operating temperature could be different from the motor temperature at starting. The controller resistance might correspond to the operating temperature that is justifiable for operation in the parameter-uncompensated system.
- (ii) If the drive system has parameter adaptation and if its performance is poor, then the estimated stator resistance might at times be higher than the actual stator resistance, leading to instability.

This section presents a solution: to track the stator resistance, so that the performance degradation and a possible instability problem can be avoided. A signal proportional to stator-resistance change is developed by using the error between the reference and actual stator current phasor. A simple proportional–integral (PI) controller is used in the stator-current phasor feedback loop to adapt the stator resistance instrumented in the controller. An analytic expression to evaluate the stator current command from the torque and stator flux linkage commands [35] is used here. The performance of the controller is shown, with dynamic simulation results for a wide variety of operating conditions, including flux-weakening.

Instability Due to Parameter Mismatch A mismatch between the controller-set stator resistance and its actual value in the machine can create instability, as is shown in Figure 8.14. This figure shows the simulations for a step stator resistance change from 100% to 80% of its nominal value at 0.5 s, with a rated torque command applied at 0.1 s. The drive system becomes unstable. This could be reasoned out as follows: As the motor resistance decreases in the machine, its current increases for the same applied voltages, which increases the flux and electromagnetic torque. The controller has the opposite effect: the increased currents, which are inputs to the system, cause increased stator-resistance voltage drops in the calculator, resulting in lower flux linkages and electromagnetic torque estimation. They are compared with their command values, giving larger torque and flux-linkages errors, resulting in the commanding of larger voltages, causing larger currents, all leading to a run-off condition.

The instability result of a step change is not realistic; the stator resistance does not in practice change in a step manner. A linearly decreasing stator resistance is simulated, and the performance is shown in Figure 8.15. Even for such a gradual change of stator resistance, note that the system becomes unstable. The controller-calculated electromagnetic torque and stator flux-linkages are almost equal to reference values and contrary to the real situation in the machine. But the S_T , S_λ , and S_θ signals to command the voltage vector signify an increase in voltage magnitude and a decrease in slip speed. Hence, any scheme using reference torque and flux linkages for parameter compensation would not be effective. For example, the air gap power-feedback control for parameter compensation in the indirect vector-controlled drive system is successful, but it will not work in this drive system.

The parameter mismatch between the controller and the machine also results in a nonlinear relationship between torque and its reference, making it an imperfect

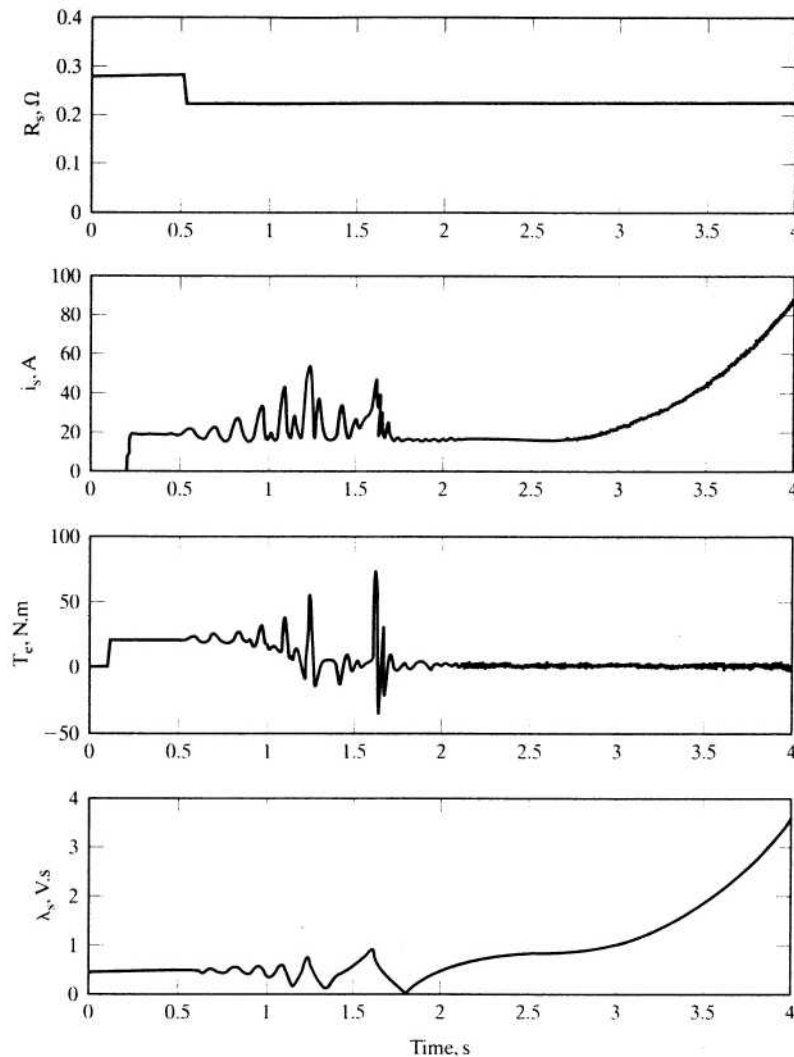


Figure 8.14 Instability due to parameter mismatch for a step stator-resistance change

torque amplifier. This will have undesirable consequences in a torque drive and to a smaller extent in the speed-controlled drive systems. The motor-resistance adaptation is essential to overcome instability and to guarantee a linear torque amplifier in the direct torque-controlled drive. A stator-resistance parameter adaptation scheme to achieve these objectives is described in the next section.

Stator-Resistance Compensation

A. Scheme A block-diagram schematic of the stator-resistance compensation scheme is shown in Figure 8.16; its incorporation in the drive schematic is given in

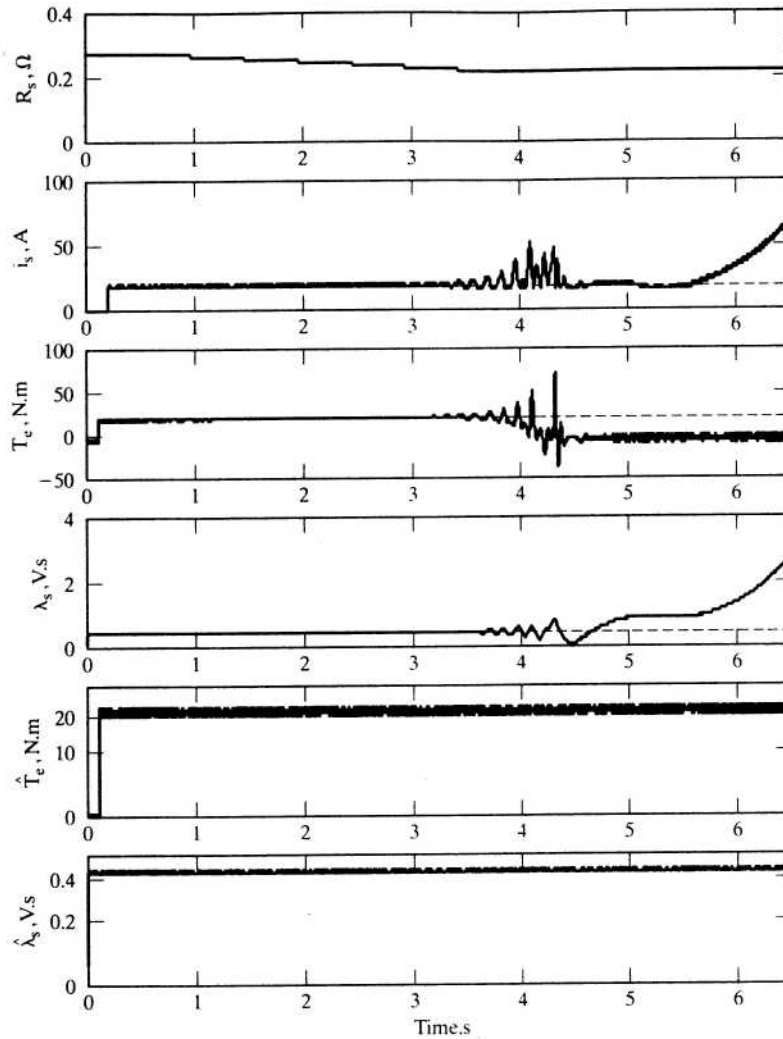


Figure 8.15 Instability due to parameter mismatch for a linearly decreasing stator-resistance change

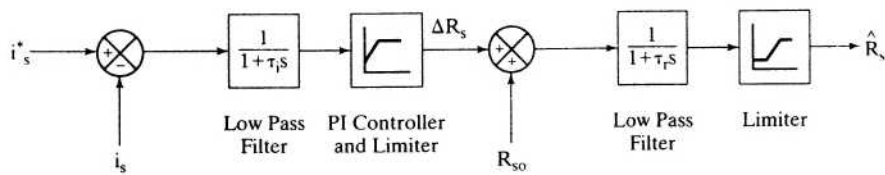


Figure 8.16 Block-diagram schematic of the adaptive stator-resistance compensator

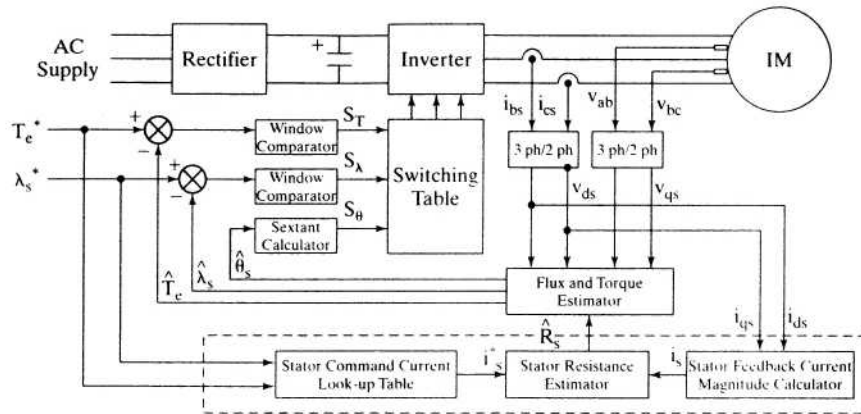


Figure 8.17 Stator-resistance adaptation in the DTC drive

Figure 8.17 and shown in dotted lines. This technique is based on the principle that the error between the measured stator-feedback current-phasor magnitude i_s and its command i_s^* is proportional to the stator-resistance variation.

The incremental stator resistance for correction is obtained through a PI controller and limiter. The current error is processed through a low-pass filter that has very low cutoff frequency, in order to remove high-frequency components contained in the stator-feedback current. The low-pass filter does not generate any adverse effect on the stator-resistance adaptation if the filter time constant is chosen to be smaller than the adaptation time constant. Incremental stator resistance, ΔR_s , is continuously added to the previously estimated stator resistance, R_{s0} . The final estimated value, $\hat{R}_s$, is obtained as the output of another low-pass filter and limiter. This low-pass filter is necessary for a smooth variation of the estimated resistance value. The final signal is the updated stator resistance and can be used directly in the controller. The above algorithm requires the stator current-phasor command for implementation. An analytic procedure to compute the stator current command from the torque and stator flux-linkages commands is presented in reference [35].

B. Dynamic Simulation Results Dynamic simulations are performed to validate the performance of the torque-controlled drive system with the compensation technique. The induction motor and drive system details used in the simulation are given next:

$$\begin{aligned} &5 \text{ HP, } 200 \text{ V, } 60 \text{ Hz, } 3 \text{ Phase, Y Connection, } 4 \text{ Poles, } 1766.9 \text{ rpm, } R_s = 0.277 \, \Omega, \\ &R_r = 0.183 \, \Omega, L_s = 0.0553 \text{ H, } L_r = 0.056 \text{ H, } L_m = 0.0538 \text{ H, } J = 0.01667 \text{ kg-m}^2, \\ &B = 0 \text{ N-m-s/rad, Rated } \lambda_s = 0.433 \text{ V-s.} \end{aligned}$$

The proportional and integral gains of the adaptation controller are 0.01 and 0.0005, respectively. The stator-resistance sampling time is 5 ms, and the cutoff frequencies for the current and resistance low-pass filters are 2 and 0.24 Hz, respectively.

In the implementation of the drive algorithm, only six nonzero voltage vectors are used, and two zero voltage vectors are excluded. It is noted that this has no impact on the basic performance of the system. Because the stator-resistance voltage drop is a significant portion of the applied voltage at low speed, the performance of the system

at low speed, from variation in the stator resistance, deteriorates much more compared with the performance at high speed. Therefore, the rotor speed is limited to 0.1 p.u. in the simulations, to illustrate the performances in the most affected region.

Figure 8.18 shows the simulation results for a step change in stator resistance in a parameter-uncompensated torque-drive system and in a compensated system. The system controller has the nominal value of stator resistance; after 0.5 s, stator resistance is changed to twice its nominal value, and the corresponding effects are studied. Rated

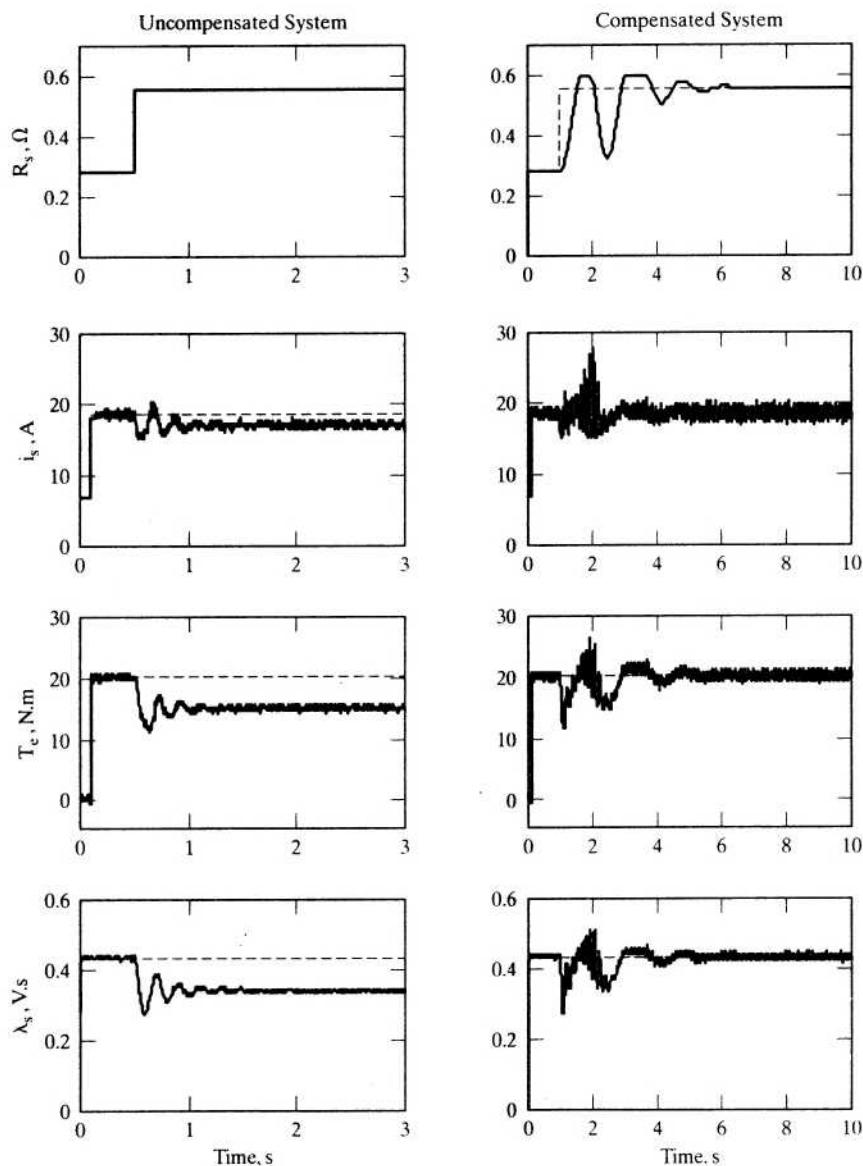


Figure 8.18 Step response for parameter-uncompensated and compensated systems

torque command is applied at 0.1 s after the stator flux linkage has reached the steady state. Parameter adaptation is initiated after 0.2 s. In the uncompensated system, right after the change of resistance, the generated electromagnetic torque decreases by nearly 25% in steady state, with much higher drop during the transient. The stator flux linkages and stator current show similar worsening. In the compensated system, it is noticed that stator-resistance estimate suffers in the initial transient state and converges gradually to its final actual value in steady state. All the other variables also have initial transient state but reach their final values in steady state. A step variation in the stator resistance is rather an extreme test and not a significant case encountered in practice.

In actual operating conditions, the rate of change of temperature is very slow and so is that of the stator resistance. Figure 8.19 covers this situation. Stator

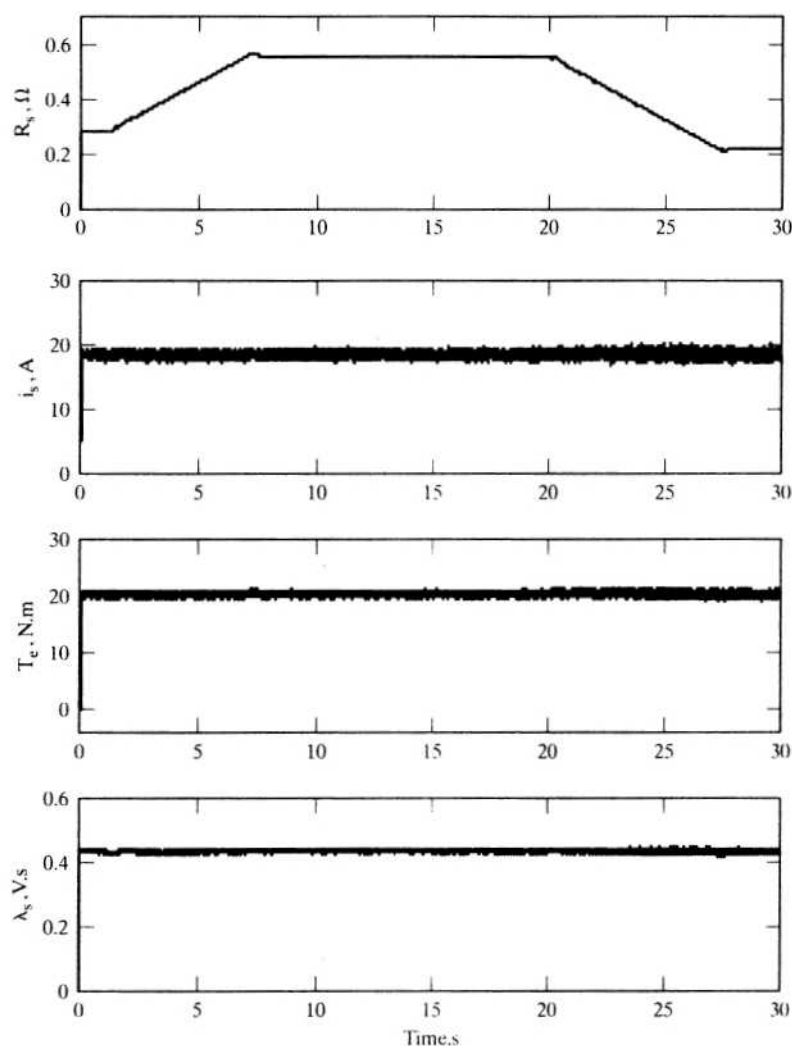


Figure 8.19 Ramp response for a parameter-compensated system

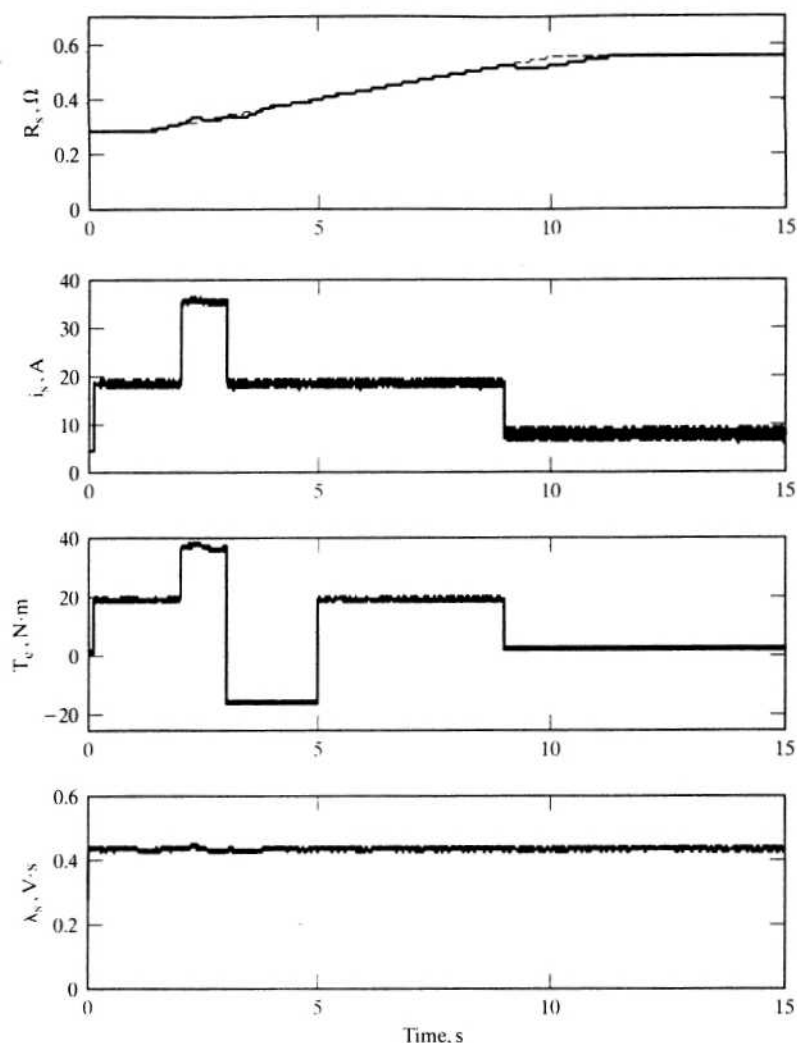


Figure 8.20 Ramp response for a parameter-compensated system

resistance is increased linearly from nominal value to twice its nominal value in 7 s, then decreased linearly from twice to 0.8 times its nominal value. The stator-resistance estimate, stator current, torque, and stator flux linkages are tracking the references very closely and experience no oscillation or transient during tracking. Note that the instability problem encountered with no parameter adaptation is solved by using this adaptive compensation scheme.

Step changes in torque command from 1 to 2 p.u. and then from 2 p.u. to -1 p.u. while the stator resistance is being ramped up have no adverse effects on the drive system, as is shown in Figure 8.20. The flux linkages hardly vary during these

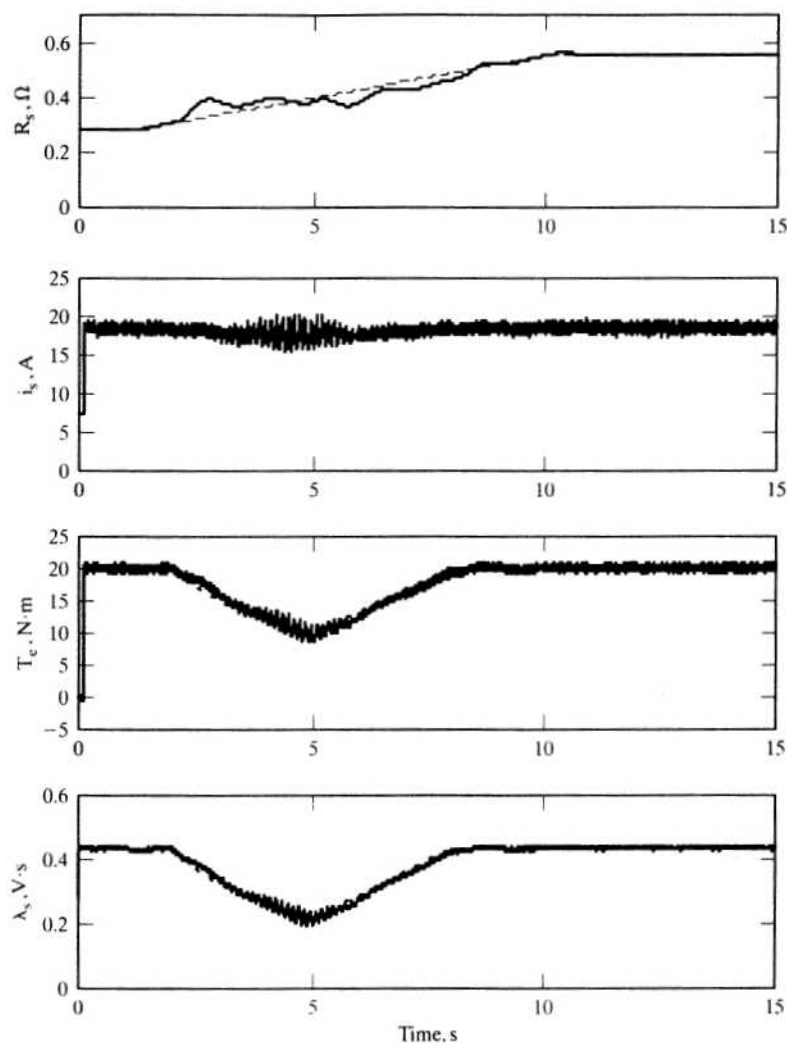


Figure 8.21 Ramp response in field-weakening mode

changes, and torque linearity is perfectly maintained between the torque and its command in this adaptation scheme.

The dynamic performance in the flux-weakening range is shown in Figure 8.21. The stator resistance is ramped from nominal value to twice its nominal value. Stator flux-linkages command and torque command are proportionally decreased from and increased linearly again to original reference values. The tracking of motor variables and stator resistance is achieved with hardly any transients, thus proving the effectiveness of the adaptive controller in the flux-weakening region also.

8.4 DERIVATION OF INDIRECT VECTOR-CONTROL SCHEME

In this section, the indirect vector controller is derived from the dynamic equations of the induction machine in the synchronously rotating reference frames. Reference could be made to the derivation of dynamic equations of the induction machine in Chapter 5. To simplify the derivation, a current-source inverter is assumed. In that case, the stator phase currents serve as inputs; hence, the stator dynamics can be neglected. In turn, that leads to omitting the stator equations from further consideration.

The rotor equations of the induction machine containing flux linkages as variables are given by

$$R_r i_{qr}^e + p \lambda_{qr}^e + \omega_{sl} \lambda_{dr}^e = 0 \quad (8.27)$$

$$R_r i_{dr}^e + p \lambda_{dr}^e - \omega_{sl} \lambda_{qr}^e = 0 \quad (8.28)$$

where

$$\omega_{sl} = \omega_s - \omega_r \quad (8.29)$$

$$\lambda_{qr}^e = L_m i_{qs}^e + L_r i_{qr}^e \quad (8.30)$$

$$\lambda_{dr}^e = L_m i_{ds}^e + L_r i_{dr}^e \quad (8.31)$$

In these equations, the various symbols denote the following: R_r , the referred rotor resistance per phase; L_m , the mutual inductance per phase; L_r , the stator-referred rotor self-inductance per phase; i_{dr}^e and i_{qr}^e , the referred direct and quadrature axes currents, respectively; and p , the differential operator d/dt . ω_{sl} is slip speed in rad/sec, ω_s is electrical stator frequency in rad/sec, ω_r is electrical rotor speed in rad/sec, and λ_{dr}^e and λ_{qr}^e are rotor direct and quadrature axes flux linkages, respectively.

The resultant rotor flux linkage, λ_r , also known as the rotor flux-linkages phasor, is assumed to be on the direct axis, to reduce the number of variables in the equations by one. Moreover, it corresponds with the reality that the rotor flux linkages are a single variable. Hence, aligning the d axis with rotor flux phasor yields

$$\lambda_r = \lambda_{dr}^e \quad (8.32)$$

$$\lambda_{qr}^e = 0 \quad (8.33)$$

$$p \lambda_{qr}^e = 0 \quad (8.34)$$

Substituting equations (8.32) to (8.34) in (8.27) and (8.28) causes the new rotor equations to be

$$R_r i_{qr}^e + \omega_{sl} \lambda_r = 0 \quad (8.35)$$

$$R_r i_{dr}^e + p \lambda_r = 0 \quad (8.36)$$

The rotor currents in terms of the stator currents are derived from equations (8.30) and (8.31) as

$$i_{qr}^e = -\frac{L_m}{L_r} i_{qs}^e \quad (8.37)$$

$$i_{dr}^e = \frac{\lambda_r}{L_r} - \frac{L_m}{L_r} i_{ds}^e \quad (8.38)$$

Substituting for d and q axes rotor currents from equations (8.37) and (8.38) into equations (8.35) and (8.36), the following are obtained:

$$i_f = \frac{1}{L_m} [1 + T_r p] \lambda_r \quad (8.39)$$

$$\omega_{sl} = K_{it} \left[\frac{L_r}{T_r} \right] \left[\frac{T_e}{\lambda_r^2} \right] = K_{it} R_r \left[\frac{T_e}{\lambda_r^2} \right] = \frac{L_m}{T_r} \frac{i_T}{\lambda_r} \quad (8.40)$$

where

$$i_f = i_{ds}^e \quad (8.41)$$

$$i_T = i_{qs}^e \quad (8.42)$$

$$T_r = \frac{L_r}{R_r} \quad (8.43)$$

$$K_{it} = \frac{2}{3} \frac{2}{P} \quad (8.44)$$

The q and d axes currents are relabeled as torque- (i_T) and flux-producing (i_f) components of the stator-current phasor, respectively. T_r denotes the rotor time constant. The equation (8.39) resembles the field equation in a separately-excited dc machine, whose time constant is usually on the order of seconds. Likewise, that the induction-motor rotor time constant is also on the order of a second is to be noted.

Similarly, by the same substitution of the rotor currents from (8.37) and (8.38) into the torque expression, the electromagnetic torque is derived as

$$T_e = \frac{3}{2} \frac{P}{2} \frac{L_m}{L_r} (\lambda_{dr}^e i_{qs}^e - \lambda_{qr}^e i_{ds}^e) = \frac{3}{2} \frac{P}{2} \frac{L_m}{L_r} (\lambda_{dr}^e i_{qs}^e) = K_{te} \lambda_r i_{qs}^e = K_{te} \lambda_r i_T \quad (8.45)$$

where the torque constant K_{te} is defined as

$$K_{te} = \frac{3}{2} \frac{P}{2} \frac{L_m}{L_r} \quad (8.46)$$

Note that the torque is proportional to the product of the rotor flux linkages and the stator q axis current. This resembles the air gap torque expression of the dc motor, which is proportional to the product of the field flux linkages and the armature current. If the rotor flux linkage is maintained constant, then the torque is simply proportional to the torque-producing component of the stator current, as in the case of the separately-excited dc machine with armature current control, where the torque is proportional to the armature current when the field current is constant. Similar to the dc machine armature time constant, which is on the order of a few

milliseconds, the time constant of the torque current is proved to be also on the same order in a later section and is equal to the stator-transient time constant. The rotor flux linkages and air gap torque equations given in (8.40) and (8.45), respectively, complete the transformation of the induction machine into an equivalent separately-excited dc machine from a control point of view.

The stator-current phasor is the phasor sum of the d and q axes stator currents in any frames; it is given by

$$i_s = \sqrt{(i_{qs}^e)^2 + (i_{ds}^e)^2} \quad (8.47)$$

and the dq axes to abc phase-current relationship is obtained from

$$\begin{bmatrix} i_{qs}^e \\ i_{ds}^e \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta_f & \cos\left(\theta_f - \frac{2\pi}{3}\right) & \cos\left(\theta_f + \frac{2\pi}{3}\right) \\ \sin \theta_f & \sin\left(\theta_f - \frac{2\pi}{3}\right) & \sin\left(\theta_f + \frac{2\pi}{3}\right) \end{bmatrix} \begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix} \quad (8.48)$$

compactly expressed as

$$i_{qd} = [T][i_{abc}] \quad (8.49)$$

where

$$i_{qd} = [i_{qs}^e \ i_{ds}^e]^t \quad (8.50)$$

$$i_{abc} = [i_{as} \ i_{bs} \ i_{cs}]^t \quad (8.51)$$

$$[T] = \frac{2}{3} \begin{bmatrix} \cos \theta_f & \cos\left(\theta_f - \frac{2\pi}{3}\right) & \cos\left(\theta_f + \frac{2\pi}{3}\right) \\ \sin \theta_f & \sin\left(\theta_f - \frac{2\pi}{3}\right) & \sin\left(\theta_f + \frac{2\pi}{3}\right) \end{bmatrix} \quad (8.52)$$

where i_{as} , i_{bs} , and i_{cs} are the three phase stator currents. Note that the elements in the T matrix are cosinusoidal functions of electrical angle, θ_f . The electrical field angle in this case is that of the rotor flux-linkages phasor and is obtained as the sum of the rotor and slip angles:

$$\theta_f = \theta_r + \theta_{sl} \quad (8.53)$$

and the slip angle is obtained by integrating the slip speed and is given as

$$\theta_{sl} = \int \omega_{sl} dt \quad (8.54)$$

From these derivations, a drive scheme and its phasor diagram are developed.

8.5 INDIRECT VECTOR-CONTROL SCHEME

A vector controller accepts the torque and flux requests and generates the torque- and flux-producing components of the stator-current phasor and the slip-angle, θ_{sl} , commands. The request/command values and the controller-instrumented parameters are denoted with asterisks throughout this text. From equations (8.39), (8.40), and (8.44), the commanded values of i_r^* , i_f^* , and ω_{sl} are

$$i_T^* = \frac{T_e^*}{K_{te}\lambda_r^*} = \frac{T_e^*}{\lambda_r^*} \frac{L_r^*}{L_m^*} \left(\frac{2}{3}\right) \left(\frac{2}{P}\right) = K_{it} \left(\frac{T_e^*}{\lambda_r^*}\right) \left(\frac{L_r^*}{L_m^*}\right) \quad (8.55)$$

$$i_f^* = (1 + T_r^* p) \frac{\lambda_r^*}{L_m^*} \quad (8.56)$$

$$\omega_{sl}^* = K_{it} \left[\frac{L_r^*}{T_r^*} \right] \left[\frac{T_e^*}{(\lambda_r^*)^2} \right] = K_{it} R_r^* \frac{T_e^*}{(\lambda_r^*)^2} = \frac{L_m^*}{T_r^*} \frac{i_T^*}{\lambda_r^*} \quad (8.57)$$

The command slip angle, θ_{sl}^* is generated by integrating ω_{sl}^* . The torque-angle command is obtained as the arctangent of i_T^* and i_f^* . The field angle is obtained by summing the command slip angle and rotor angle. With the torque- and flux-producing components of the stator-current commands and rotor field angle, the qd axes current commands (and, hence, abc phase-current commands) are obtained as follows. The transformation from the flux- and torque-producing currents to the qd axes current commands are derived from the phasor diagram shown in Figure 8.1. The relevant steps involved in the realization of the indirect vector controller are as follows:

$$\begin{bmatrix} i_{qs}^* \\ i_{ds}^* \end{bmatrix} = \begin{bmatrix} \cos \theta_f & \sin \theta_f \\ -\sin \theta_f & \cos \theta_f \end{bmatrix} \begin{bmatrix} i_T^* \\ i_f^* \end{bmatrix} \quad (8.58)$$

and

$$\begin{bmatrix} i_{as}^* \\ i_{bs}^* \\ i_{cs}^* \end{bmatrix} = [T^{-1}] \begin{bmatrix} i_{qs}^* \\ i_{ds}^* \\ 0 \end{bmatrix} \quad (8.59)$$

where

$$T^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 1 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 1 \end{bmatrix} \quad (8.60)$$

By using equations (8.58) to (8.60), the stator q and d axes and abc current commands are derived as

$$\begin{aligned} i_{qs}^* &= i_s^* \sin \theta_s^* \\ i_{ds}^* &= i_s^* \cos \theta_s^* \\ i_{as}^* &= i_s^* \sin \theta_s^* \\ i_{bs}^* &= i_s^* \sin \left(\theta_s^* - \frac{2}{3} \pi \right) \\ i_{cs}^* &= i_s^* \sin \left(\theta_s^* + \frac{2}{3} \pi \right) \end{aligned} \quad (8.61)$$

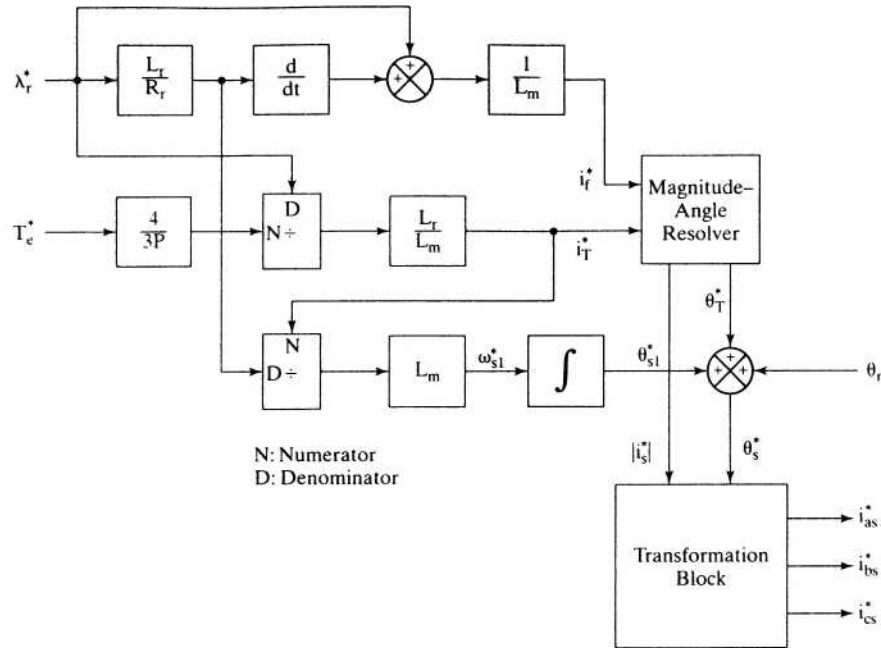


Figure 8.22 Functional block diagram of a current-source indirect vector controller

where

$$\theta_s^* = \theta_r + \theta_T^* = \theta_r + \theta_{sl}^* + \theta_T^* \quad (8.62)$$

The current commands are enforced by the inverter with suitable control techniques outlined elsewhere. A realization of the scheme in block-diagram form is shown in Figure 8.22. The flowchart for the generation of control variables in the indirect vector-control scheme is given in Figure 8.23. An implementation of the indirect vector-control scheme on an inverter-fed induction motor is discussed in the next section.

8.6 AN IMPLEMENTATION OF AN INDIRECT VECTOR-CONTROL SCHEME

An implementation of the indirect vector-controlled induction motor is shown in Figure 8.24. The torque command is generated as a function of the speed error signal, generally processed through a PI controller. The flux command for a simple drive strategy is made to be a function of speed, defined by

$$\begin{aligned} \lambda_r^* &= \lambda_b; & 0 \leq \pm \omega_r \leq \pm \omega_{\text{rated}} \\ &= \frac{\omega_b}{|\omega_r|} \lambda_b; & \pm \omega_b \leq \pm \omega_r \leq \pm \omega_{r(\text{max})} \text{ and } |\omega_r| \geq \omega_b \end{aligned} \quad (8.63)$$

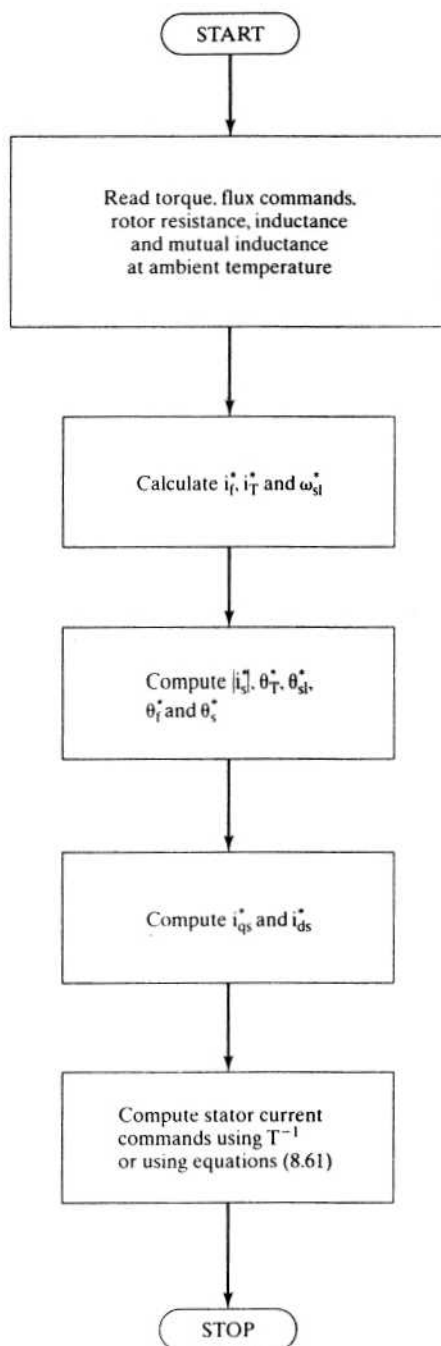


Figure 8.23 Flowchart for the indirect vector-controlled induction motor

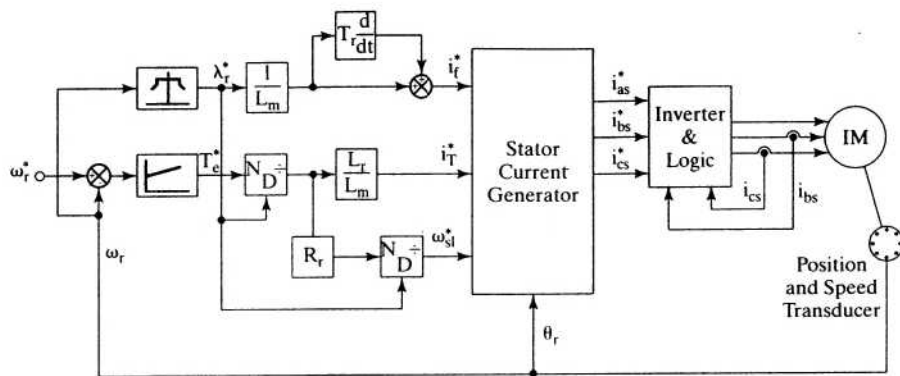


Figure 8.24 The implementation of the indirect vector-controlled induction motor servo

where λ_b and ω_b are the rated or base rotor flux linkages and rotor speed, respectively. The flux is kept at rated value up to rated speed; above that, the flux is weakened to maintain the power output at a constant, very much as in the dc motor drive. In such a case, the rotor flux-linkages programming is a complex task; it is considered in detail in section 8.10. By following the flow chart given in Figure 8.15, the three-phase stator-current commands are generated. These current commands are simplified through a power amplifier, which can be any standard converter-inverter arrangement discussed in the earlier chapters. For small motor drives (of less than 25-hp rating), the power amplifier is of the type shown in Figure 8.8. The regenerative energy is dumped in the brake resistor in this power circuit. The output of the inverter is connected to the induction-motor stator terminals. The third-phase current is reconstructed from the two measured phase currents if the system does not have a grounded neutral.

The rotor position, θ_r , is measured with an encoder/synchronous resolver and converted into necessary digital information for feedback. Some transducers are currently available to convert the rotor position information into velocity; they can be used to eliminate a tachogenerator to obtain the velocity information. The controllers are implemented with microprocessors; a vast amount of literature is available for reference on this topic. Next in order are the tuning considerations for the indirect vector controller.

Example 8.3

Prove that the transfer function between the rotor flux linkage and its command is unity; from that, prove that the transfer relationship between the torque and its command is unity in the indirect vector controller for the induction machine. Assume that the machine and controller parameters match and that there is no time delay between the currents and their commands.

Solution From the flux-linkage equation derived from the rotor equation, the rotor flux linkage is given as

$$\lambda_r(s) = L_m \frac{i_f(s)}{(1 + sT_r)}$$

but the flux-producing component of the stator current, i_f , is equal to its command value, i_f^* , because there is no time delay and the vector controller enforces the command faithfully through the inverter, which then is written as

$$i_f = i_f^*$$

The flux-current command obtained from the vector controller is given by

$$i_f^*(s) = \frac{(1 + sT_r^*)}{L_m^*} \lambda_r^*(s)$$

From these equations, the flux-producing component of the stator-current phasor is substituted in terms of the flux linkage command and motor parameters as

$$\lambda_r(s) = \frac{L_m}{L_m^*} \frac{(1 + sT_r^*)}{(1 + sT_r)} \lambda_r^*(s)$$

As the motor and controller parameters match, the rotor flux linkages and its command become equal, giving the transfer function between the rotor flux linkage and its command as unity. Then the transfer relationship between the electromagnetic torque and its command is derived as follows:

$$T_e = K_{te} \lambda_r i_T = K_{te} \lambda_r i_T^* = K_{te} \lambda_r \frac{T_e^*}{K_{te}^* \lambda_r^*}$$

As the motor and controller parameters match, the torque constant for the controller and the machine become the same, and the ratio between the rotor flux linkage and its command is unity. That makes the torque equal to its command value, resulting in a unity transfer relationship between them.

Example 8.4

Derive the time constants associated with the flux- and torque-producing channels in an indirect vector-controlled induction motor drive.

Solution

- (i) Flux channel: From the equation of the rotor flux linkages, it is seen that the time constant of the rotor flux-linkages channel is the rotor time constant, denoted by T_r . Its value is on the order of seconds, like the field constant in the separately-excited dc machine.
- (ii) Torque channel: To derive the time constant in this channel, it is assumed that the rotor flux-linkages phasor is maintained constant. Then the torque is influenced only by the torque-producing component of the stator-current phasor, i_T . Therefore, the time constant associated with this channel will be the time constant involved in the transfer function between this current and the corresponding voltage, which is the quadrature axis voltage in synchronous reference frames. It is derived as follows.